

DEEP LEARNING MODELS TO MAP DEFORESTATION BASED ON SENTINEL 1 COHERENT FEATURES IN THE SOUTHERN BORDER OF AMAZON

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Abstract:

The Amazon is the largest continuous area of tropical forest in the world. It is under pressure from deforestation, degradation, wildfires, and the expansion of agriculture, livestock, and illegal mining. Existing long-term optical and radar monitoring services with high precision are being developed, with products like Land Use and Land Cover (LULC) and deforestation alerts. However, they are effective only in the dry season, and some require a lot of human labor to generate information about forest disturbance. To improve the knowledge about the LULC and deforestation process, specifically developing algorithms applied in the Amazon in three different landscapes, this research contributes to elaborate deep learning (DL) classifiers that employ Synthetic Aperture Radar (SAR) with coherent attributes to deliver minimum mapping unit (MMU) at 30 m and revisit time of 12 days. The U-Net, Semantic Flow (SF-Net), and Long Short-Term Memory (LSTM) deep learning models were adapted to use SAR images from Sentinel 1. These three DL models were improved to include blocks such as residual learning, dilated convolutional, attention mechanisms, and squeeze-and-excitation, which were fine-tuned through trials in the Optuna framework. The model Sentinel 1 scene 54 622/627, U-Net model, and 4 classes reached the highest overall accuracy and intersection over-union (IoU) in

order of 0.95 and 0.66, respectively. The less precise mapping was noticed by Sentinel 1 scene 83 617/622, LSTM model, and 4 classes with a global accuracy of 0.61 and IoU of 0.36. The deep learning model that achieved the lowest error was U-Net with an RMSE of 0.43 and standard deviation of 0.43, and it was considered a random error. On the contrary, the SF-Net and LSTM showed systematic error, which reached RMSE between 0.38 to 0.83 and a standard deviation between 56.53 to 114.69. The most precise LULC classes were provided by Forest (Fo) and Deforestation (De), which achieved the highest values of F1-Score with 0.97 and 0.92, respectively. On the opposite way, it was the Non-Forest (NF) and Water (Wa) classes that obtained a lower F1-score in order of 0.51 and 0.72, respectively. Observing the Taylor and Target diagrams, Sentinel 1 scene 83 617/622 was more suitable for deep learning modeling considering the U-Net architecture and its positions towards Ground Control Points (GCP). A new DL method was developed for Sentinel 1 and its coherent features, allowing mapping of the flat to hilly relief of the Southern Amazon during the dry season.

Keywords: Deep Learning, Synthetic Aperture Radar, Land Use and Land Cover, Deforestation, Amazon.

1. Introduction

Deforestation and forest degradation negatively affect carbon stock, biodiversity, and the provision of ecosystem services, leading to landscape vulnerability and less resilience to climate change (Ometto et al., 2022). The world's forest cover is estimated at an absolute area of 40.1 million km², which is 31% of Earth's land surface and provides direct dependence on the resources for 1.3 billion people (FAO, 2024). It is estimated that 4.2 million km² of forest cover were lost between 1990 and 2020, and the land use and land cover (LULC) have been changing under deforestation rates in the order of 102 thousand km².yr⁻¹ in the last 5 years (FAO and UNEP, 2020). The highest deforestation rates are observed in Africa, followed by South America and Asia.

There are several initiatives with reliable results to monitor the processes of LULC and deforestation in the world, such as optical-based systems like the Global Land-Cover Classification (Zhang et al., 2021), Global Forest Change (Hansen et al., 2013a)(Hansen et al., 2013b), Global Land Cover Facility (Song et al., 2018). The more recent mapping services are forthcoming with highly detailed spatial resolutions with 5 up to 30 m and high performances of global accuracy between 0.59 to 0.98 that are Deter by Diniz et al., (2015), ESA World Cover by Zanaga et al. (2022) and Global Watch Forest by CTREE (2024), and also, the promising radar-based systems like CESBIO/TropiSCO by Mermoz et al. (2021), the RADD by Reiche et al. (2021) and JJ-Fast by Watanabe et al. (2021) with performances of global accuracy between 0.26 to 0.91. These long-term optical monitoring services are majorly based on Terra, Aqua, Amazonia, CBERS, Landsat, Sentinel 2, and PlanetScope, and the radar-based monitoring services use Sentinel 1 and Also-2. They have been achieving high precision in detecting small disturbances in the forest with minimum mapping unit (MMU) in the order of 0.1 to 25 ha, quick revisit times of 5 to 42 days, and coming up with several classes (2 up to 30) like deforestation and degradation. In addition, there are new geospatial tools to recognize the regrowth of natural forests (Hoekman et al., 2020). However, they are more effective in the dry season (June up to October), they have difficulty mapping close to the Equator line (around -4 to +4 degrees), in complex landscapes with steeper slopes or flooded plains, and some of them demand extensive human labor to generate training data.

Amazon is the largest continuous area of tropical forest in terms of worldwide coverage and biodiversity (FAO, 2020), which has estimated ecosystem services with a maximum contribution of around 57 to 737 USD.ha⁻¹.yr⁻¹ in forest remnants (Strand et al., 2018). The Project for Monitoring Deforestation in the Legal Amazon by Satellite (PRODES) by INPE

(2024) is the major monitoring service for Amazon that has been continuously measuring deforestation by visual interpretation for 36 years in an approximated area of 5 million km², it indicates a total deforestation of 850.5 thousand km², which is lost proportionally to 16.96% of the Brazilian Legal Amazon (INPE, 2024). Recently, Mapbiomas by Souza et al. (2020) handed in the yearly LULC maps for the entire Brazil using optical images and semi-automated classification. Although Brazilian institutional efforts culminated in drastic reductions by 2012, reaching a deforestation rate of 4,571 km².yr⁻¹, recently, the deforestation rate increased in 2022 to 11,594 km².yr⁻¹, which is very far from the target established in the Paris Agreement by 2020, that rate was 3,925 km².yr⁻¹ (MMA, 2022).

Amazon demands special attention due to the growing pressure exerted by demographic, economic, and social changes, especially by the expansion of agriculture and livestock, new roads, energy projects, and spontaneous or subsidized migrations (Barreto et al., 2008). The significance of the Amazon lies in its social, economic, and ecosystem aspects, and on the other hand, the challenges of monitoring and surveillance are in its territorial extension, severe cloudiness, and landscape complexity (Ab'Saber, 2002). The main commitment of Brazil is to achieve the reduction of deforestation rates in forested areas which collaborates with life on land (fifteenth sustainable goal development from United Nations).

Synthetic Aperture Radar (SAR) can capture ground images using your own illumination with almost all-day and all-weather conditions using penetrability through clouds, forest canopy, soil, and ice depending on the wavelength. In addition, the radar that was designed for coherent sampling allows Earth Observation (EO) based on persistent monitoring which consists of a complex signal of amplitude and phase at predictable interference (Sullivan, 2004; Woodhouse, 2017). The new SAR systems can achieve hourly temporal resolutions and spatial resolutions as fine as 0.2 m, and are capable of providing data radar interferometry, polarimetry, radargrammetry, and tomography (Ouchi, 2013; Paradella et al., 2012). For SAR applied to forests, it is essential to use longer wavelengths and polarimetric data to penetrate the vegetation, assess its vertical structure, and classify the scattering mechanisms using coherent indexes and polarimetric decompositions (Estimates, 2021; Lee and Pottier, 2017; van Der Sanden, 1997). Accurate forest disturbance mapping lies in choosing the radar system, its acquisition geometry, and derivative features from the backscattering. It is also crucial to understand the environmental variables and the techniques suitable for classifying the SAR data.

There have been more than 30 new SAR satellites in the last 40 years, most of them operated by North America, Europe, and Japan, and this number is increasing each year with new lighter payloads, tandem arrangement, better spatial resolutions, swath widths, and imaging modes (Ouchi, 2013; Song et al., 2024). Initiatives such as Copernicus have ascertained over three decades of accurate C-band SAR data with global coverage and the same acquisition geometry (ESA, 2024a). Future research should exploit this ever-increasing wealth of SAR satellites to address pressing issues in the fields of geoscience and the environment.

The usage of SAR data over forests has contributed to several findings such as: i) speckle noise reduction should retain the texture in heterogeneous areas and the non-linear behavior in forests (Dasari et al., 2022; Dong et al., 2016; Kang et al., 2023; Quegan and Yu, 2001); ii) optical-SAR fusion provides the best performance over forested environments, but it is challenging in frequently cloud-covered areas (Reis et al., 2020; Tavares et al., 2019; Vaglio Laurin et al., 2013); iii) Bayesian techniques, time series and recurrent models have the best potential for forest monitoring, but often require many separability investigation, dense time series and precise auxiliary information like masks, thematic maps, and digital surface models (DSM) (Ballère et al., 2021; Doblas et al., 2020; Koyama et al., 2019; Mermoz et al., 2021; Nicolau et al., 2021; Reiche et al., 2018); iv) various SAR observables and products such as backscatter intensity, texture, interferometric coherence, topography, and polarimetric

parameters improve forest mapping performance, but typically require strong temporal correlation (hour up to 6 days) and high spatial resolution (1 to 6 m) which have constraints to reach large minimum mapping unit (1 to 10 ha) or limited spatial extent of the study (Altarex et al., 2023; Guimarães et al., 2018; Jin et al., 2014; Liesenberg and Gloaguen, 2013; Sugimoto et al., 2022; Watanabe et al., 2021); v) biomass retrieval models have been used to estimate carbon stocks using advanced techniques like interferometry, polarimetry, and tomography, but these studies must cover wider forest landscapes to be capable of mapping higher amounts of biomass and varieties of tree species, and their respective disturbances. The biomass modeling requires detailed forestry inventories and high precision on the canopy height provided by DSM. (Narvaes et al., 2023; Persson and Huuva, 2024; Pourshamsi et al., 2021; Quegan et al., 2019; Soja et al., 2021a, 2018); and finally, vi) recently developed classifiers based on conventional statistical approach, machine learning (ML) and deep learning (DL) perform well in terms of accuracy and minimum mapping unit, but many challenges remain due to complex landscapes, seasonality, and significant requirements on in situ (sampling) and remote sensing data (radiometric and geometric errors) (Fu et al., 2017; Kuck et al., 2021; Maretto et al., 2021; Matosak et al., 2022; Mullissa et al., 2023; Reis et al., 2020; Solórzano et al., 2021; Tavares et al., 2019; Zhao et al., 2022). The current techniques for mapping forest disturbances shall be focused on achieving transferability, across locations, minimum mapping units, and remote sensing datasets.

The DL algorithms are changing EO which outcomes in massive results in remote sensing providing high accuracies, improved models, large areas mapped, and detailed change detections. Since SAR images have higher ambiguity in the backscattering, it is essential to exploit the spatial context and the time series of the amplitude signal instead of usage only the statistical behavior (Ma et al., 2019, 2017). In addition, several SAR studies corroborate using coherent features to improve classifier performance (Parikh et al., 2020; Passah et al., 2022; Shafique et al., 2022; Zhou et al., 2016). There are still challenges to overcome in DL applications such as self-supervised training, handling complex numbers, overfitting, class imbalance, and vanishing or exploding gradients.

It is mandatory to amplify the spatial-temporal context on Amazon, which means increasing the dataset with annotations. There may be increased uncertainty and decreased assertiveness in recognizing targets due to Amazon's landscape and its environment. The development of ML/DL classifiers demands knowledge of state-of-the-art techniques from computer sciences, but it is crucial to understand remote sensing besides explainable geophysical conditions. (Joshi et al., 2016; Maxwell et al., 2018; Shafique et al., 2022; Tuia et al., 2025). The opportunity here is given by the Sentinel 1 in C-band dual-polarization from (ESA, 2024a) in its better update frequency of 12 days and 20 m of spatial resolution that was used to discriminate LULC classes and subtle changes in the Amazonian landscape.

This study proposed to map LULC in the Amazon focused on deforestation. It employed a Sentinel 1 SAR process chain and implemented DL models for training and inference based on intensity, interferometric coherence, and polarimetric decomposition. It is supposed that the classifiers' errors vary with each Sentinel 1 scene, which presents different landscapes. In this sense, Sentinel 1 scenes in the southern Amazon were mapped and evaluated under deforestation pressure during the dry season.

2. Study Area

Southern Amazon has been chosen as the study area (Figure 1) because it is under pressure as a new hotspot of deforestation with 35% of all new alerts in the Amazon, which is driven by the advance of livestock and construction of highways (Chaves et al., 2024). Also, the location is between -5° to -12° of latitude which increases the chances to obtain Sentinel 1 and 2 in co-orbital arrangement without cloud for training and evaluation of the models. The

study area was defined by 6 scenes of Sentinel 1 that were split into three path/frames composed of 2 scenes in the descending orbit. The three path/frames of the study area together cover an area of 810 km × 350 km and are located at the southern border of the Amazon, which recently had high rates of deforestation and wildfires (INPE, 2024; MapBiomas, 2024).

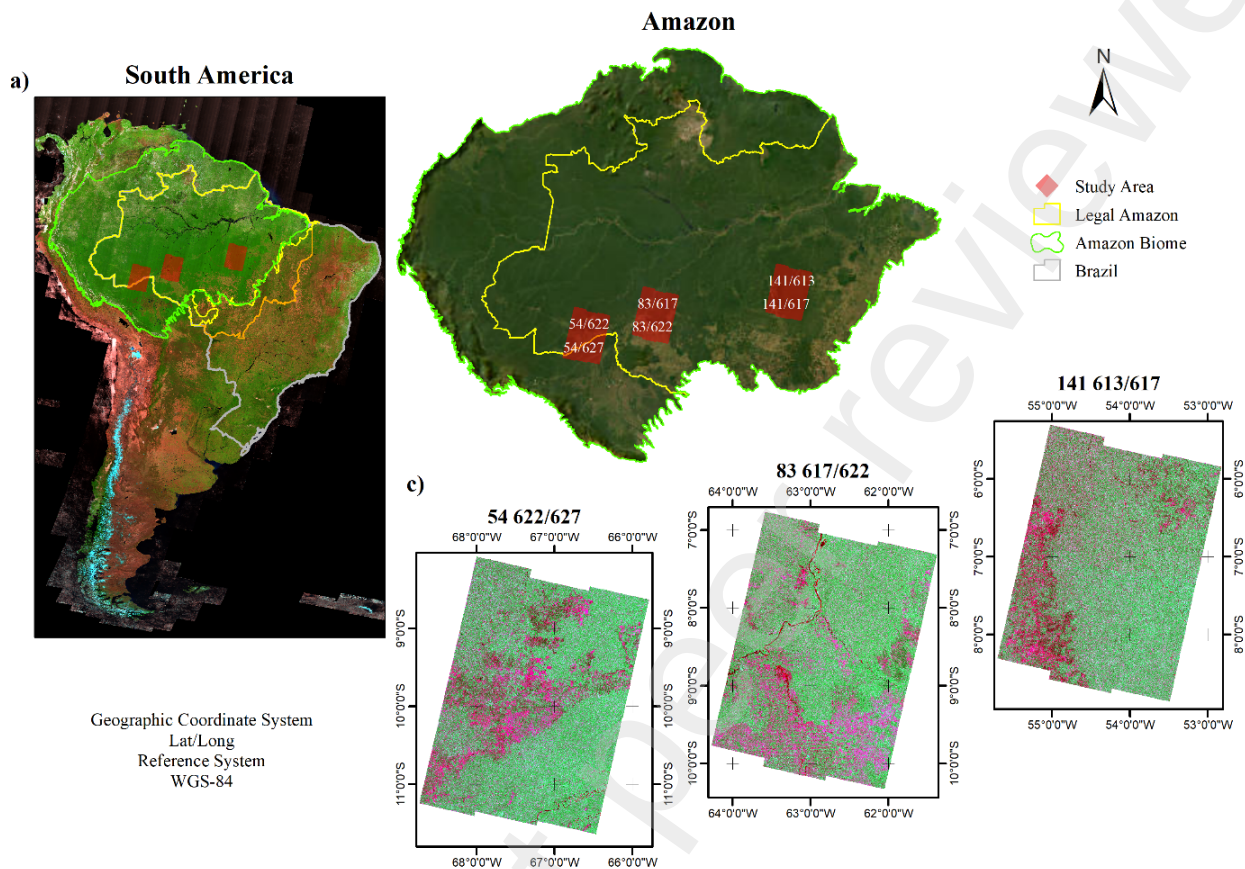


Figure 1 - a) The study area is located in South America; b) the Amazon biome and its Legal Amazon (Brazilian territory) observed by the Sentinel 1 acquisition scenario available by (ESA, 2024a); c) the three path/frames of the Sentinel 1 descending orbit located on the southern of Amazon.

There are three main roads in the study area called BR-319, BR-230, and BR-163 which cross in the SW-NE and S-N directions and are the main vectors of regional population and economy. Despite the area of interest having protected areas of 49.70%, the pressure of deforestation occurs along the roads and in the surroundings. The relief and slopes vary significantly from flat, undulated, plateau, cliffs up to hilly areas where the altimetric quota is between 48 to 752 m, and the slope average of 7.88°, both auxiliary information observed in the Shuttle Radar Topography Mission (SRTM). The environmental characteristics of the three path/frames, like LULC and landscape description, are in Table 1.

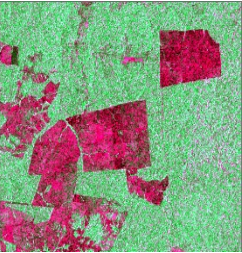
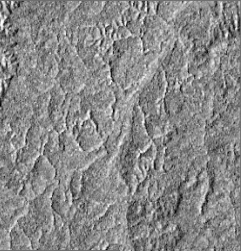
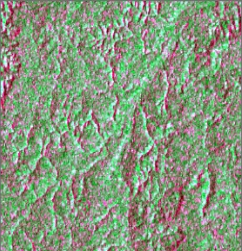
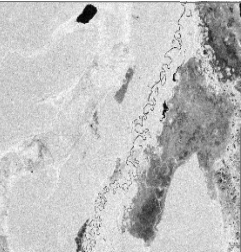
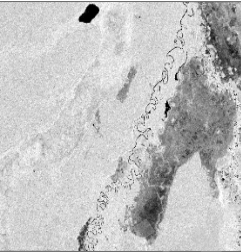
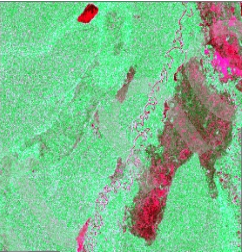
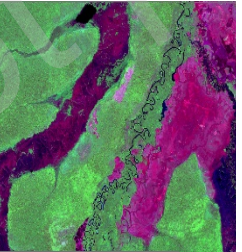
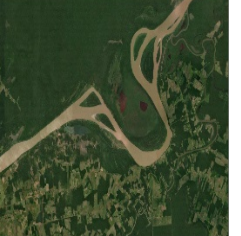
Table 1 - It describes the environmental features for the three path/frames in the study area.

Environmental Characteristics		Sentinel 1 Path/Frame ¹		
		54 622/627	83 617/627	141 613/617
Date	Training Inference	23/07/2022	01/07/2022	20/07/2022
		24/07/2024	08/06/2024	29/07/2024
Area (km ²)		87,077.61	69,757.42	85,458.23
Alerts ¹ (%)		0.48	0.61	0.49
Deforestation ¹ (%)		24.88	35.71	14.09
Forest ¹		73.42	57.01	84.66
Non-Forest ¹		0.84	5.06	0.80
Water ¹		0.87	2.22	0.46
Geology ²	Major Units	Iça, Solimões, Mutum-Paraná formations	Jamari and Nova Monte Verde complex, Iça, Palmeiral and Roosevelt formations, intrusive suites	Xingu complex, Aruri, Salustiano and Triunfo, formations, Beneficente, Gorortire and Iriri groups, intrusive suites
	Minor Units	alluvial fans and terraces, detrital-lateritic cover	alluvial fans and terraces, detrital-lateritic cover	alluvial fans and terraces, detrital-lateritic cover
Geomorphology ²	Major Units	Amazon plain and depression, basin and deposits sedimentary	Amazon plain and depression, residual plateau, basin and deposits sedimentary	Amazon plain and depression, residual plateau, basin and deposits sedimentary
	Minor Units	residual plateau, floodplain, unconsolidated sedimentary deposits	floodplain, terraces, unconsolidated sedimentary deposits	Cachimbo plateau, Pardo ridge, floodplain, unconsolidated sedimentary deposits
Soil ²	Major Units	yellowish red, yellow and red latosol	yellowish red, yellow and red latosol, ultisol, spodosols	ultisol yellowish red, yellow and red latosol,
	Minor Units	gleisoi, plinthosols	gleisoi, plinthosols, leptosols, clay loam	gleisoi, plinthosols, leptosols, clay loam
Vegetation ²	Major Units	lowlands ombrophilous forest with bamboo/palm trees	ombrophilous, open submontane, and bush forests	dense submontane and ombrophilous forest
	Minor Units	submontane and alluvial forests and meadow with grass	savannah, meadow, lowlands, alluvial forests and grass	lowlands and alluvial forests
Average Altitude ³ (m)		186.5	128.69	259.85
Average Slope ³ (°)		3.33	2.48	5.82
Rain ⁴ (mm.month ⁻¹)	Rainy	264.80	270.85	551.98
	Dry	54.67	45.53	80.37

Source: ¹ ESA (2024b), INPE (2024), ²IBGE (2012, 2020, 2020), ³USGS (2015) ⁴INMET (2024)

The LULC mapping has focused on describing Deforestation, Forest, Non-Forest and Water, following the method stated by (INPE, 2024). From the optical and SAR images, these four LULC classes were described according to their physical coverage using visual interpretation based on color, tonality, texture, form, and context, as presented in Table 2.

Table 2 - The key visual interpretation to recognize the Amazon LULC classes that were observed on the sigma nought (σ^0), interferometric coherence (γ), and the Cloude-Pottier dual-pol decompositions with Entropy ($DPol_H$) and Alpha ($DPol_\alpha$).

LULC	Sentinel 1		Sentinel 2		PlanetScope	SAR Recognition	
	VH	VV	RCov GMin BGrad	RSwir GNir BRed	RRed GGreen BBlue	Elements and Patterns	Features
Deforestation (De)						Color: black, dark to moderate gray Tonality: very low to medium Texture: very smooth to smooth, slightly heterogenous, homogenous Shape: geometric, regular, strips, elongated to compact patches Context: borders well defined, bare soil with forest, connected with previous deforestation	σ^0 : very weak to weak signal, specular and diffuse scattering, low penetrability, polarized targets γ : partially coherent to coherent, moderate to high Pol_H : slight to moderate randomness, partially pure targets Pol_α : low, slight high, specular and surface backscattering
Forest (Fo)						Color: dark, light and very light gray Tonality: medium to high Texture: moderate to strong rough, heterogenous Shape: irregular, large and random geometries, contiguous patches Context: large areas, borders well defined with rivers and deforestation, controlled by relief	σ^0 : moderate to strong signal, penetrability, volumetric scattering, depolarized targets γ : uncoherent, very weak Pol_H : moderate to severe randomness, partially to high distributed targets Pol_α : intermediate, surface and volumetric backscattering
Non-Forest (NF)						Color: dark and light gray Tonality: low to medium Texture: smooth to moderate rough, homogeneous Shape: irregular, patches, large, random, contiguous Context: random variations in large areas, borders well defined with rivers, patches controlled by relief	σ^0 : very weak to moderate signal, depolarized targets, intermediate penetrability γ : partially coherent, weak to moderate Pol_H : low to moderate randomness, partially distributed targets Pol_α : low to intermediate, surface to volumetric backscattering
Water (Wa)						Color: black to dark gray Tonality: very low to low Texture: very smooth, slightly heterogenous, homogenous Shape: irregular, elongated, dendritic, meandric, linear to curve, narrow to wide geometries Context: lines, patches, well defined borders, controlled by relief	σ^0 : very weak to weak signal, polarized targets, specular scattering, not penetrating γ : uncoherent, very weak Pol_H : low randomness, pure targets Pol_α : low, slight high, specular and surface backscattering

3. Method

This research uses three deep neural network classifiers with supervised learning approaches. Two of them are based on convolutional neural networks (CNN): U-Net, and Semantic Flow (SF-Net), while one is based on Recurrent Neural Networks (RNN): Long-Short-Term Memory (LSTM). These algorithms have elaborated to train and infer the LULC classes based on SAR backscattering intensity, interferometric coherence, and Cloude-Pottier dual-polarized decomposition products.

The procedures can be synthesized in: i) sampling strategy was used to obtain Ground Control Points (GCP) from the PlanetScope images and use them to elaborate a Random Forest (RF) LULC classification for the entire path/frame based on Sentinel 1 and 2 in co-orbital arrangement (Chatziantoniou et al., 2017; Tavares et al., 2019); ii) SAR processing chain for the estimation of backscattering intensity, coherence, and polarimetric attributes was developed based on ESA (2024c), Ferretti et al. (2007), Guimarães et al. (2017, 2018, 2020), Lee and Pottier, (2017b) and SARMAP (2016); iii) DL classification pipelines consisted of the U-Net (Maretto et al., 2019; Ronneberger et al., 2015), SF-Net (Li et al., 2024, 2020) and LSTM (Ienco et al., 2017; Matosak et al., 2023, 2022) models, which mapped the Amazon environments following some previous studies as Altarez et al. (2023), Jin et al. (2014), Liesenberg et al. (2016) and Whyte et al. (2018); and iv) thematic accuracy assessment used the confusion matrix, F_1 -score, dice score and intersection over-union (IoU), statistical tests, Shapley Additive exPlanations (SHAP), and Taylor and Target diagrams (Congalton and Green, 2019; Lundberg and Lee, 2017; Taylor, 2001). The procedures conducted in this research are shown below in Figure 2:

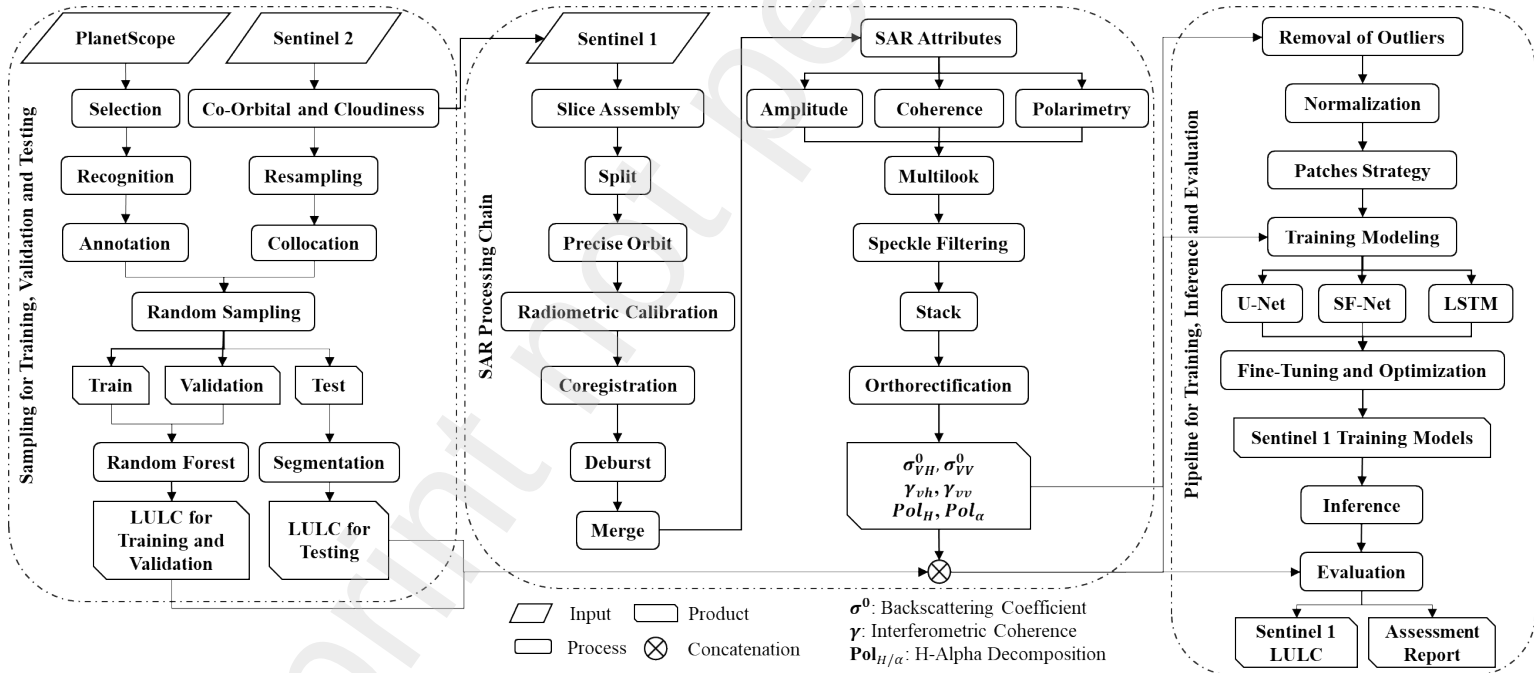


Figure 2 - The workflow shows the three main procedures carried out: i) the sampling strategy with optical images from Sentinel 2 and PlanetScope for training and evaluation of the LULC maps; ii) the SAR processing chain used to obtain backscattering and the coherent features; and iii) the pipeline to training and finetuning the models and inferring the LULC predictions.

The procedures in the workflow were carried out in Python scripts considering several geospatial libraries to handle and pre-process the datasets, PyTorch and Optuna frameworks to design and fine-tune the DL models, and the SNAP Toolbox platform which processed the images and generated the features (ESA, 2024b). The script implementation used the Anaconda

environment and Spyder graphic interface. The main repositories for coding were DeepGeo, Satellite Image Deep Learning, and AI4EO, all of them available on GitHub. All models were carried out in an RTX 3080 GPU with 16 GB memory.

3.1 Sampling for Training, Validation and Testing

The manual mapping with GCP was the primary source of reliable ground information observed in PlanetScope images that were provided by NICFI (2024) and Planet Labs PBC (2024), and also, Sentinel 2 provided by (ESA, 2024a). These optical datasets were critical to generating samples for training, fine-tuning during the validation, and assessing the DL model's performance during the testing. PlanetScope was obtained at 3A level (orthorectified and atmospheric corrected) with 4 spectral bands in the visible and near-infrared, and 4.77 m of the ground sample. Sentinel 2 was acquired in Level 2A (atmospheric corrected) with 13 bands in the visible, near-infrared, and shortwave infrared with different spatial resolutions as 10, 20, and 60 m.

Three random stratified disproportional samplings were carried out to obtain 44,984 points split into 50% for train, 25% for validation, and 25% for testing. Sentinel 2 had a critical role in improving the continuous sampling, which provided the spectral bands resampled to 10 m and was collocated with Sentinel 1, adding detailed spatial context and dimensionality in the input channels. The sampling strategy followed the assumptions of randomness, independence, good spatial distribution, and an error expected at least three times more accurate than the data to be tested. Both optical images allowed us to achieve better remote sensing parameters in recognizing LULC classes, given the spectral and spatial resolution compared to Sentinel 1 which has 2 channels and spatial resolution at 20 m (Olofsson et al., 2014).

The SAR image was defined by the availability criteria in the co-orbital arrangement between Sentinel 1 and 2, the shortest timespan with 3 days, less than 2% cloudiness, and during the dry season. This procedure was useful for achieving continuous sampling over SAR data and RF classification, as stated by Chatziantoniou et al. (2017) and Tavares et al. (2019).

Despite several new DL approaches, there are very ML techniques capable of proceeding decently to map LULC with Sentinel 1 and 2 (Joshi et al., 2016; Maxwell et al., 2018). Among them was the RF is an ensemble learning algorithm that applies the bootstrap aggregation to sample random subsets in a decision tree with replacement, the subsets ensembled are combined through major voting with the noisy samples set in a specific tree called out-of-bag (Belgiu and Drăgu, 2016; Breiman, 2001). It was considered 15,000 GCP per Sentinel 1 path/frame, the bootstrapped approach had 65% for training and the out-of-bag had the remaining 35% for cross-validation. The number of trees in the forest (Ntree) was chosen as 1000 and the number of variables at each node of the tree (Mtry) was 5 (square root of input channels), and the maximum depth was 300, according to Guimarães et al. (2020) and Tavares et al. (2019).

The continuous samples given by RF were significantly important to the DL pipelines when they passed through down-sampling kernels, providing a detailed spatial context. The RF classifier consumed the training and validation GCP, and the Sentinel 1 and 2 images to yield a continued LULC map which conceded samples in the DL pipeline. On the other hand, the testing GCPs were associated with segments to assess the DL models as a cross-validation dataset. The multiresolution segmentation has set parameters such as 15 of scale, 0.1 of shape, and 0.5 compacity.

The GCP and the RF LULC maps are shown in Figure 3. The RF classification had previous accuracies between 0.94 and 0.97, and the seven most important features were shortwave infrared, near-infrared, Red, VV channel, VH channel, interferometric coherence, and isotropic decomposition. The GCPs were collected in proportion to 33.34% of Deforestation, 33.34% of Forest, 13.33% of Non-Forest, and 19.98% of Water.

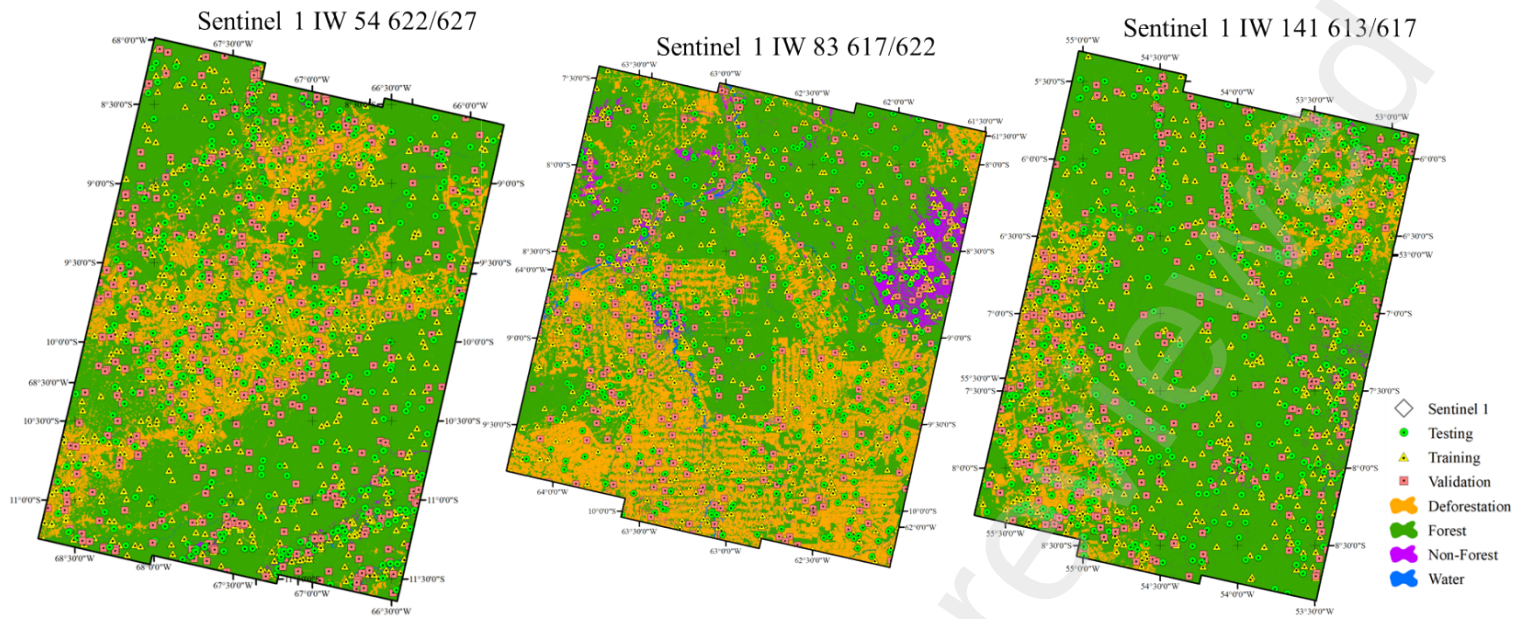


Figure 3 - It shows the spatial distribution of the GCP for training, validation, and testing where they were obtained through the PlanetScope images. Afterward, the GCP and the Sentinel 1 and 2 in the co-orbital arrangement were used to classify the LULC in the three path/frame by RF.

3.2 SAR Processing Chain

Sentinel 1 processing chain has been carried out to ensure a comparable and stable signal over time and its coherent features, as well as precise image alignment and coverage in the same path/frame following a previous study by Guimarães et al. (2020). The parameters of the SAR chain are shown in Table 3 with Equations (1), (2), and (3) to obtain the amplitude, coherence, and polarimetric decomposition, respectively. The SAR images were geocoded using a pixel spacing at 30 m, and the cartography projection as Lat/Long and WGS-84 reference system.

The dimensionality of the Sentinel 1 images was increased by the interferometric coherence and H- α dual polarimetric decomposition, according to previous studies such as Canisius et al. (2019) and Shuai et al. (2019). Coherence measures the similarity of the target, it considers the complex cross-correlation by the amplitude and phase of each pixel in the master and slave images through a moving window operator, it varies between [0, 1] where 1 means high coherent behavior observed in the image pair. H- α polarimetric decomposition explains the randomness (H) and type of the scattering (α). The entropy (H) means if the signal is homogenous with 0 value or heterogenous with 1 value, low entropy indicates a dominant scatter as a pure signal (isotropic behavior) like a double-bounce target, and oppositely, high entropy signifies a random mixture of scatterers like a forest target. The alpha (α) describes the mechanism of scattering such as surface in around 0° values, volumetric and di-pole behavior with intermediate values of 45°, and double bounce in around 90° values.

Table 3 - The parameters applied in the SAR process chain and features obtained from Sentinel 1.

SNAP's Tools ¹	Parameters	Attribute
Slice Assembly Apply Orbit Files	2 Adjacent Frames in the Path Sentinel Precise	
Calibrate	Sigma Nought; Complex Output	$\begin{cases} \sigma_{vh}^0 = \frac{P_s}{P_i} \frac{A_\sigma}{A_\sigma} \\ \sigma_{vv}^0 = \frac{P_s}{P_i} \frac{A_\sigma}{A_\sigma} \end{cases} \quad (1)$
Split Back Geocoding Enhanced Spectral Diversity Deburst Merge	IW 1, IW 2, IW 3; 18 Bursts Copernicus DEM 30 m; 21-Points Truncate Coherence 0.3 VV, VH IW 1, IW 2, IW 3; VV, VH	
Coherence Estimation	8 x 2 Window Size	$\begin{cases} \gamma_{vh} = \frac{\langle x_1 x_2^* \rangle}{\sqrt{\langle x_1 ^2 \rangle \langle x_2 ^2 \rangle}} \\ \gamma_{vv} = \frac{\langle x_1 x_2^* \rangle}{\sqrt{\langle x_1 ^2 \rangle \langle x_2 ^2 \rangle}} \end{cases} \quad (2)$
Polarimetric Matrix	T2 Matrix	
Polarimetric Decomposition	Refined Lee; H- α Dual-Pol	$\begin{cases} Pol_H = -\sum_{i=1}^3 p_i \log_3(p_i); p_i = \frac{\lambda_i}{\sum_{k=1}^3 \lambda_k} \\ Pol_\alpha = \sum_{i=1}^3 p_i \alpha_i \end{cases} \quad (3)$
Multilook Speckle Filtering Stack	8 x 2 Window Size; Intensity Gamma Map; 3 x 3 Master; Orbit	
Terrain Correction	Copernicus DEM 30 m; Bilinear; 30 m	

Source: ¹ ESA (2024c). Where σ^0 is complex radar signal calibrated and projected on the ground range as sigma nought for two polarizations ($\sigma_{vh}^0, \sigma_{vv}^0$). P_s and P_i are the scattered and incident power respectively, and they are usually related with the digital number, and also A_σ is the reference area projected in the ground range that is usually given by a constant; γ is complex cross-correlation applied on both polarizations that estimates the conjugated mean in each pixel $\langle x_1 x_2^* \rangle$, and the root square of the standard deviation $\sqrt{\langle |x_1|^2 \rangle \langle |x_2|^2 \rangle}$ by a moving window; $Pol_{H/\alpha}$ are the features based on Cloud-Potier dual-polarization decomposition, p_i is the pseudo-probability of the eigenvalue and λ_i is the eigenvalue in the possible set λ_k which they are given by the coherence matrix (T2).

3.3 DL Pipelines for Training and Inference

The DL pipelines have been developed focusing on data generation, models, training loops, fine-tuning optimization, and inference. 18 DL models based on 3 architectures, 3 Sentinel 1 paths/frames, and 2 arrangements of LULC classes were elaborated.

The dataset generation handled the loading and preprocessing procedures for the input images and the annotations. The load considered the images as their original scale (32-bit float),

height, width, number of channels, and geolocation (EPSG, Geodetic Parameter Dataset), which were extracted from the big geotiff file (.tif). In the same way, the annotations were identified and rasterized from the shapefiles (.shp) that were composed of single polygons with a specific column attribute with the LULC class. A geospatial function was applied in the annotations to read, rasterize, resample, and verify the alignment with the input images.

The input images provided a statistical description, and if "no data" and "not a number" values were observed, they were removed from the dataset. After that, the datasets were recapped by the 99th percentile, safeguarding the absence of outliers that are common in Sentinel 1 in the edges or related to thermal noise (ESA, 2024b).

Normalization has been applied to the recapped data to ensure stability, comparison, convergence, and consistency, which feeds and accelerates the deep learning training loop (Montavon et al., 2012). Min-max scaling was employed within values at $[0, 1]$.

The high-dimensional Sentinel 1 data outcomes in a large footprint to process at once, which is not practical due to computational constraints in this DL pipeline. To alleviate the memory footprint, the strategy was to split the Sentinel 1 image into smaller parts, which are called patches (Maretto et al., 2019). The dataset was split sequentially into patches with 64 x 64 in width and height, using functions to create overlapping patches, load images and labels, and reassemble them.

U-Net is a symmetric convolutional neural network focused on image segmentation based on encoding and decoding abstract spatial features at the pixel level (Ronneberger et al., 2015). The encoding oversees extracting features through a kernel that passes the image and recognizes shapes, textures, edges, borders, and patterns, trading off with less spatial resolution. The bottleneck is the deepest part of the model, capturing the most compressed characteristics and linking the encode and decode parts. The decoding is in charge of recovering the original spatial resolution besides retaining meaningful information, which uses skip connections to preserve the detailed spatial context and perform the prediction. The U-Net model developed is shown in Figure 4.

U-Net has been proven for almost ten years to be a well-known and effective model for LULC, although there are several advantages in exploiting advanced techniques that enhance the extraction of spatial features (Clark et al., 2023; Ramos and Sappa, 2025; Yu et al., 2021). This research employed techniques such as residual blocks in the backbone of the models and optional blocks like attention mechanisms, squeeze-and-excitation (SE), and dilated convolution. These techniques boosted the CNN models into converging and capturing spatial context from SAR images without using deeper layers.

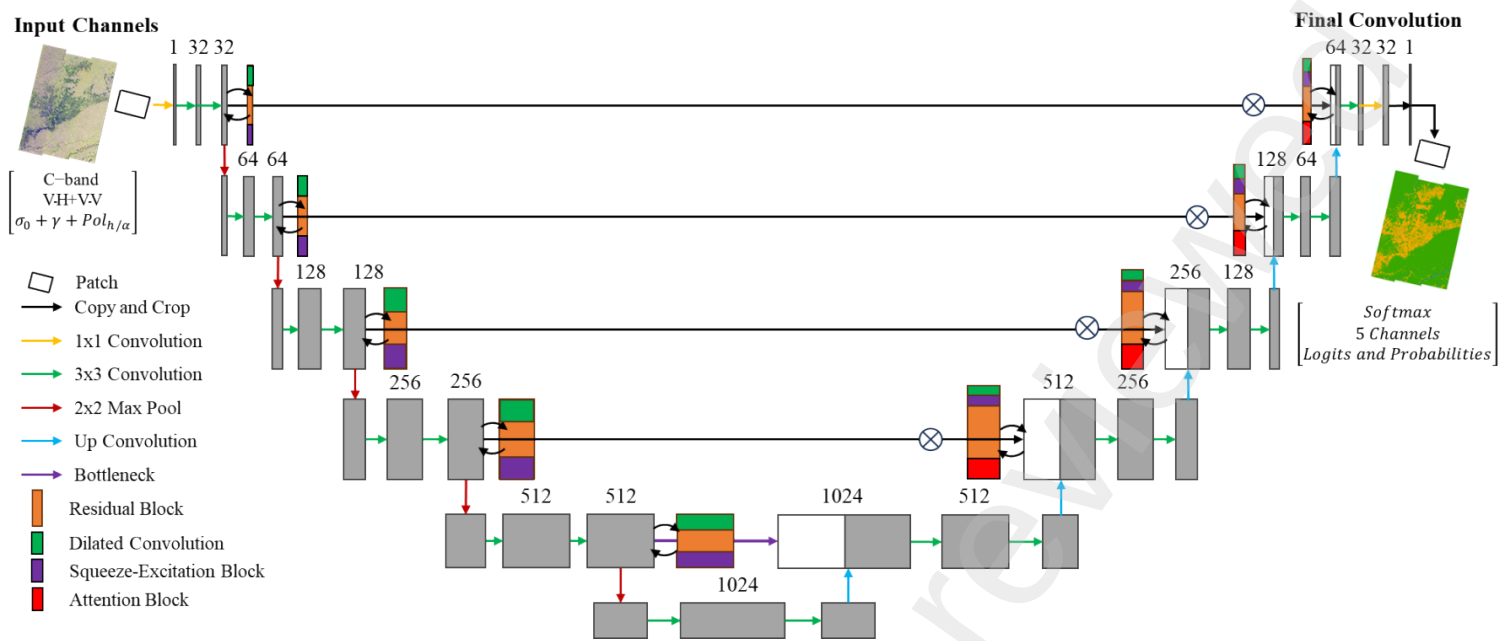


Figure 4 - In the U-Net architecture, where the input is a Sentinel 1 image with coherent features that go through the encoding and decoding path up to the final convolution, the output is a LULC map with logits and probabilities.

Residual learning is a technique that enables addressing the learning even in the condition of small gradients, and instead of learning directly from the reference map, the idea is learning from the residual map, preventing degradation of performance (He et al., 2015). The attention mechanism is employed to give high importance to specific features and spatial positions, and it is used in the decoder part as a feature map, specifically to improve skip connections and locate more relevance in some areas and less in others (Vaswani et al., 2017). The SE block enhances the power representation of the channels in the feature map. It considers the relationship between the aggregation of spatial information and learning the importance of each channel, which suppresses less relevant channels and amplifies the usage of critical ones (Hu et al., 2017). The dilated convolutional is a technique that modifies the standard kernel by creating a new one with gaps that increase the receptive field, which provides the advantage of covering a large area with the same parameters, and it means a multi-scale context preserving the spatial resolution (Yu and Koltun, 2015). A single patch composed of batch size, channels, height, and width goes through the pipeline established of convolution, normalization, dropout, residual blocks, optional blocks, max pooling (downsampling), upsampling, activation functions, and ends up in the final convolution. The last procedure performs the softmax function that takes the vector of real numbers (logits) and transforms it into a probability distribution $[0, 1]$ with a specific label.

SF-Net is a semantic segmentation method focused on fusing multi-scale features extracted from different resolutions and abstract levels. The algorithm promotes the fusing of features and semantics levels through the flow alignment module. The feature alignment is allocated like vectors at a pixel-wise level called flow field, and a competitive match occurs to move or warp the spatial information in a grid, which magnifies the solution in the feature map (Li et al., 2024, 2020). The SF-Net's approach followed the same U-Net pipeline that is given by the optional blocks and differencing just in the backbones and proper adjustments required to insert the alignment module and warp function (Figure 5).

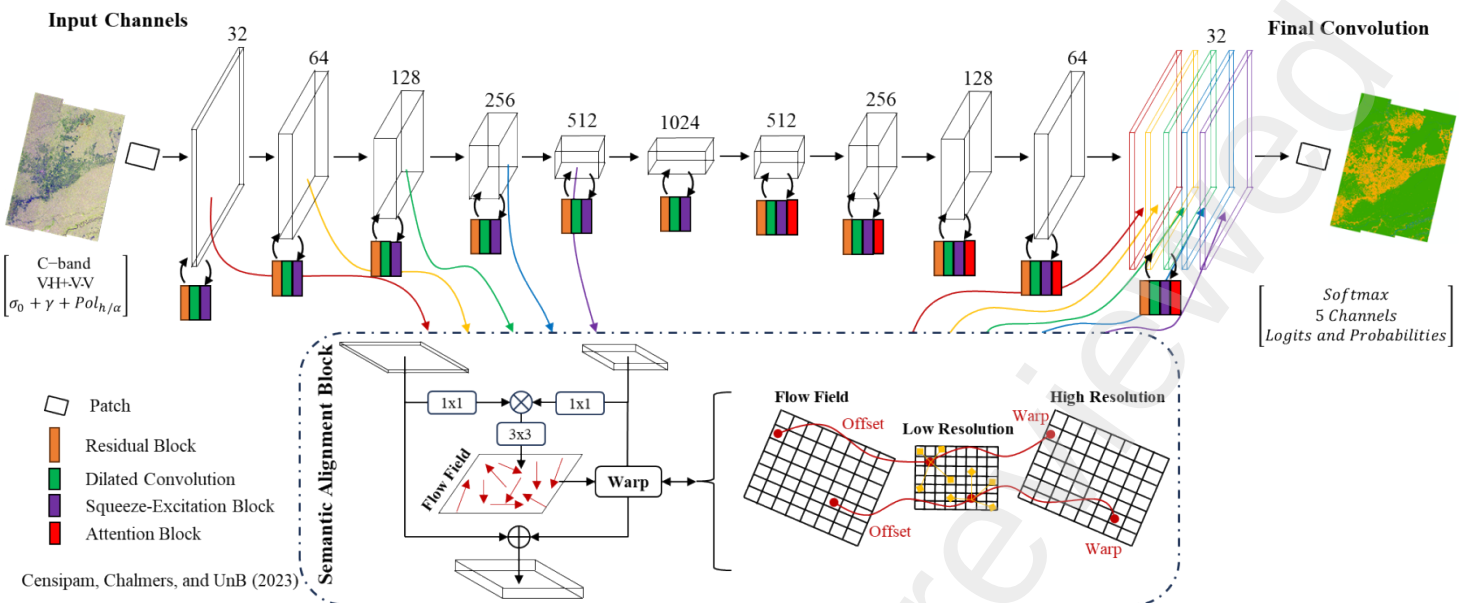


Figure 5 - The SF-Net architecture developed for Sentinel 1 and coherent attributes, the pipeline has a symmetric encoding and decoding path connected to a semantic alignment block, and the final convolution provides a LULC map with logits and probabilities.

RNN Algorithms such as LSTM have been achieving satisfactory results in mapping and detecting changes in LULC as assured by Ienco et al. (2017), Matosak et al. (2023), and Zhu et al. (2021). The LSTM is appropriate for getting around the problem of long-term dependence between observations separated by a long period. The LSTM strategy applies to a gated mechanism to regulate the flow of information in forward and backward directions that provides flexibility to retain, update, and discard abstractions (Greff et al., 2015). It has the potential for multi-class change detection and forest disturbance (Liu et al., 2025; Zhu et al., 2024). Some disadvantages are related to the high demand for memory, difficulty in fine-tuning, and common overfitting.

A short-term series was used in this research, composed of 3 dates of Sentinel 1 and their coherent features. In addition, the LSTM was designed to use advanced techniques like residual learning and optional blocks such as attention mechanism, and SE, with the exception compared to CNN models was the absence of a dilated convolutional block and the addition of a bidirectional block. The LSTM has developed a temporal modeling layer and 4 dense layers, and each dense layer has a normalization layer, a regularization layer (dropout), a non-linear activation (ReLU), and then it goes to the final dense layer with a softmax function to estimate probabilities and the label (Figure 6).

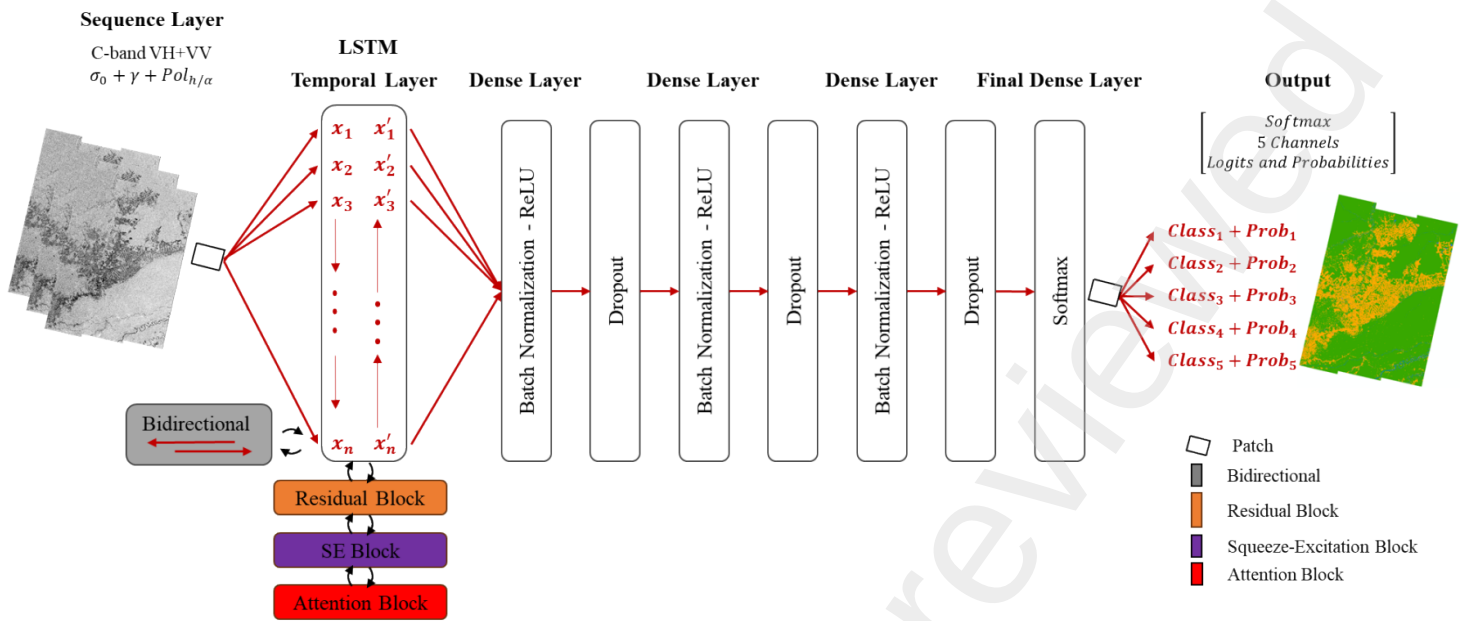


Figure 6 - The LSTM architecture developed for Sentinel 1 in a time series and their coherent attributes, the pipeline concatenates the input channels to pass through the temporal modeling and dense layers up to the final dense layer that provides a LULC map with logits and probabilities.

The LSTM pipeline starts with the patch defined and a concatenation of the input channels composed of batch size, sequence length, and feature size after the data goes through the temporal modeling layer and the fully connected dense layers. The training and retraining looping were set similarly as for CNNs, considering structure, parameters, and hyperparameters.

The set of hyperparameters directly affects the performance of the DL model, and the practical definition mostly relies on empirical experiences (Diaz et al., 2017). An objective function was defined by Optuna due to the several hyperparameters to manage and the need to optimize them based on the IoU. Optuna is a sequential model-based optimization that has approached on a Bayesian probabilistic algorithm, in this case, has set the Tree-structured Parzen Estimators (Akiba et al., 2019). This procedure consisted of 25 trials with 10 epochs each in the training loop phase, and after that, a retraining loop phase was performed on the 3 top models ranked by IoU criteria with 50 epochs each. The parameters and hyperparameters have been set and investigated by the Optuna, in which there are equal parts of all three models, and other parts are different and have categorical or numeric choices presented in Table 4.

Table 4 - The parameter and hyperparameters have been set for the U-Net, SF-Net, and LSTM.

Parameters	Values	
	U-Net and SF-Net	LSTM
Training Loop	25 Trials, 10 Epochs	
Retraining Loop	50 Epochs	
Batch Size	[64, 128, 256]	
Base Channels	[32, 144]	
Dropout	[0.1, 0.5]	
Learning Rate	[0.000001, 0.01]	
Weight Decay	[0.000001, 0.01]	
Optimizer	Adaptive Moment Estimation (Adam), Root Mean Squared Propagation (RMSprop), Stochastic Gradient Descent (SGD)	
Scheduler	Step Learning (StLR), Reduce Learning on Plateau (ReLRP), Cosine Annealing Warm Restarts (CAWR)	

Loss Function	Cross Entropy (CrE), Sparse Categorical Focal Loss (Focal), Dice Loss (Dice), Focal CrE (FCrE), Dice (DCrE)			
Inputs and Outputs	10 channels, 4 to 5 Classes		15 channels, 4 to 5 Classes	
Normalization	Convolutions Layers		Dense Layers	
Architecture	Encoder	32, 64, 128, 256, 512	Temporal Modelling	3 LSTM
	Down Sampling	Max Pooling 2x2	Layers	[2, 7]
	Bottleneck	1024	Dense Layers	3
	Decoder	512, 256, 128, 64, 32	Hidden Size	[32, 144]
Activation	Up Sampling	Transpose 2x2	Bidirectional	[True, False]
	Backbone	ReLU	Sigmoid, Tanh	
	Residual	ReLU	ReLU	
	Attention	ReLU, Sigmoid, [True, False]	ReLU, Sigmoid, [True, False]	
Output	SE	ReLU, Sigmoid, [True, False]	ReLU, Sigmoid, [True, False]	
	Output	Softmax	Softmax	

3.4 Thematic Accuracy and Evaluation

The classes of LULC were defined in previous studies and current mapping projects in Brazil (INPE, 2024). However, the training samples were studied to understand the statistical behavior and the separability between the LULC classes. It was performed the coefficient of variation (CV) the number of equivalent looks (NEL), and the interquartile boxplot (Q1-Q3) to understand the dispersion, the level of averaging assuming independence, and statistical behavior, respectively. The Mann-Whitney-Wilcoxon and Levene tests were carried out to compare the average and variance of the datasets at 5% confidence respectively, and the training samples were assumed to be non-parametric behavior, considering as well, a small and unbalanced group.

Statistical tests supported understanding the thematic error through precision and accuracy, whereas models with higher biases were not considered for the separability or importance of the variables. The Jeffries Matusita distance was used to observe the binary separability between the classes of LULC, in which the values are estimated between [0, 2], values below 1 are considered partial separable, and values above 1 are more distinct as close to 2. The importance of SAR coherent features has been analyzed to ratify properly the dimensionality in the DL models, which is expected to decrease the redundancy (Rostami and Kaveh, 2021). SHAP explains the variables and predictions in the DL model which estimates a benchmark baseline without any contribution of inputs, and after that includes the features using possible combinations to observe the results. The SHAP function has been set with 30 patches randomly sampled and non-additive real numbers that are comparable only relatively. Positive values mean better predictions and negative values the opposite (Lundberg and Lee, 2017).

The evaluation of 18 DL models was carried out using training and inference accuracy, which focused on the testing dataset (reference) and predictions through the confusion matrix, IoU, and Taylor's and Target's diagrams. The confusion matrix consists of the cross-tabulation between samples of predicted (classified) and observed (reference) elements which have matched against each other to provide the matrix's counters such as true positive (TP), false negative (FN), false positive (FP) and true negative (TN). The IoU was in charge of guiding the training loop as outcome feedback about the performance. The Taylor and Target diagrams were used to oversee the similarity, precision, and accuracy of the DL models based on unbiased errors. This approach followed good practices to assess the dataset based on a benchmark with less error, PlanetScope image has higher spectral and spatial resolution (Olofsson et al., 2014).

The assessment of the DL models' performances served as an aid in choosing the DL technique better adjusted to the Amazon environments and the usage of Sentinel 1 for mapping LULC in three different landscapes. The testing dataset had 11,244 GCP from the PlanetScope images manually classified as deforestation, forest, non-forest, and water. The reference GCP,

predictions, and counters of the confusion matrix have allowed computing all thematic accuracy metrics as presented in Table 5.

Table 5 - The assessment metrics used in the DL models and their respective equations. In addition, some figures synthesize the usage and application of thematic accuracy.

Assessment Metrics	Equations	Figure
Confusion Matrix	$\begin{cases} M_{ij} = \sum_{k=1}^N 1(y_k = i \wedge \hat{y}_k = j) \\ M_{1,1} = TP; M_{1,0} = FN; M_{0,1} = FP; M_{0,0} = TN \end{cases} \quad (4)$	
Recall	$Recall = \frac{\sum TP}{\sum TP + \sum FN} \quad (5)$	
Precision	$Precision = \frac{\sum TP}{\sum TP + \sum FP} \quad (6)$	
F-Score	$F_{score} = 2 \times \frac{Recall \times Precision}{Recall + Precision} \quad (7)$	
Root Mean Squarer Error	$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^N (y_k - \hat{y}_k)^2} \quad (8)$	
Intersection Over-Union	$IoU = \frac{TP}{FP + TP + FN} \quad (9)$	
Normalized Standard Deviation	$\begin{cases} \bar{y} = \frac{1}{N} \sum_{k=1}^N y_k; \bar{\hat{y}} = \frac{1}{N} \sum_{k=1}^N \hat{y}_k \\ \sigma_y = \sqrt{\frac{1}{N} \sum_{k=1}^N (y_k - \bar{y})^2}; \sigma_{\hat{y}} = \sqrt{\frac{1}{N} \sum_{k=1}^N (\hat{y}_k - \bar{\hat{y}})^2} \\ \sigma^* = \frac{\sigma_{\hat{y}}}{\sigma_y} \end{cases} \quad (10)$	
Correlation	$\rho = \frac{\frac{1}{N} \sum_{k=1}^N (y_k - \bar{y})(\hat{y}_k - \bar{\hat{y}})}{\sigma_y \sigma_{\hat{y}}} \quad (11)$	
Normalized Bias	$B^* = \frac{\bar{\hat{y}} - \bar{y}}{\sigma_y} \quad (12)$	
Unbiased Root Mean Squarer Error	$uRMSE = \sqrt{RMSE^2 - B^2} \quad (13)$	

Source: Congalton and Green (2019), Garcia-Garcia et al. (2018). Where M_{ij} is an element in the confusion matrix given a row (i) and column (j) that match the reference and predicted annotations respectively. N is the number of samples, y_k is a sample of reference, $\hat{y}_k$ is a sample of the prediction. **Recall**, **Precision**, **F_{score}**, **RMSE**, **IoU** are metrics based on the information provided by the confusion matrix. the normalized standard deviation (σ^*) is measured by the

mean ($\bar{y}$) and standard deviation (σ_y) of reference, and the mean ($\hat{y}$) and standard deviation ($\sigma_{\hat{y}}$) of predicted. The correlation (ρ) is measured through the covariance of reference and prediction and their respective standard deviation. The σ^* and ρ are used in the Taylor's diagram. The normalized bias (B^*) is the difference between the means of prediction and reference divided by the standard deviation of reference. $uRMSE$ measures the variability between the reference and predictions after removing the bias (B). The B^* and $uRMSE$ are used in the Target's diagram.

4. Results and Discussions

4.1 Statistical Behavior, Separability of LULC and Importance of Features

Sentinel 1 has been proven to distinguish the LULC classes regarding the dual-pol backscattering ($\sigma_{vh}^0, \sigma_{vv}^0$), moderate spatial resolution (30 m), and incidence angle (35.5°), during the dry season. The Forest (Fo) class was most discernable followed by Deforestation (De), on the other hand, the Non-Forest (NF) class was the most difficult to classify. This was noticed by the statistical behavior of the backscattering and LULC classes which suggests masking or decomposing the Non-Forest class that is less distinguishable (Table 6).

The backscattering was very similar between the path/frame which had low values in VH and high values in VV and given a lot of ambiguous signals considering the LULC. Specifically, in the Water class, it was noticed the lowest values of backscattering with the highest values of standard deviation. This indicates that the VV channel is important in classifying water bodies with specular reflection. The deforestation class was the second low signal in VH and VV which is discernable in most cases when observed in the Forest and Non-Forest classes, and also, it has a homogenous behavior if compared to the variability of the water class. It is clear the similarity between the classes Forest and Non-Forest regarding the backscattering VV polarization, in contrast, the VH channel has a difference in intensity that is remarkable and must be explored to classify.

Another aspect is to observe the CV, it expresses the dispersion of the variability of the classes where low values indicated tightly cluster around the average and high values suggest heterogeneous distribution. In this study, Forest was the class more homogenous than the other ones, except in the VH channel in the path/frame 83 617/622. Water was the class more heterogeneous, specifically for the VV channel. NEL denotes the number of independent samples contributing to the signal backscattered a low value implies low signal stability with high noise and high values mean stable backscattering given by a homogenous surface. Forest again reached the highest values, which has proven less noisy and more reliable for classification. The lower values of NEL were found in Non-Forest and Water, indicating a difficulty in classifying these classes.

The distribution of the signal backscattered which is composed of similarities and discrepancies between the LULC classes, was found according to other studies based on C-band and dual polarization SAR systems. Huang et al. (2023) had similar values of backscattering for 3 classes (Farmland, Urban Area, and Forest) and it was just Water better discernable, this study used Gaofen 3 in a Monsoon tropical region. Hansen et al. (2020) studied the separability of Sentinel 1 data for just two classes and realized that Forest has in general higher values of intensity than Non-Forest, even in different regions like tropical, subtropical, or polar. Regarding short-term SAR acquisitions and focusing on mapping, it is imperative to consider better conditions of separability given by coherent SAR attributes, shallow incidence angles, and dry seasons (Guimarães et al., 2017, 2020).

The Mann-Whitney-Wilcoxon, Lavene, and Jeffries Matusita tests were used to compare, respectively, are average, variance, and distance of the LULC training datasets (Table 7). They depicted statistical similarities between the parried classes that are harder to differ in a classifier, in this case, it was Forest/Non-Forest and Deforestation/Non-Forest. Specifically for this ambiguity in the Forest and reaching statistical differences, it is possible to exploit the radar vegetation index based on polarimetric derivatives (Hu et al., 2024; Narvaes et al., 2023).

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Table 6 - Sentinel 1 dual polarization scenes were described statistically by σ_{vh}^0 , σ_{vv}^0 and LULC, at 35.5° incidence angle and during the dry season.

Statistics (dB)	54 622/627							
	De		Fo		NF		Wa	
	VH	VV	VH	VV	VH	VV	VH	VV
Min	-29.43	-25.16	-38.32	-30.31	-28.73	-24.62	-31.63	-27.08
Max	9.55	17.94	0.00	6.92	0.00	5.95	-0.07	4.24
Mean	-16.37	-10.86	-14.01	-7.87	-14.43	-8.79	-19.63	-14.85
SD	1.91	1.89	1.16	1.23	1.91	2.06	3.51	4.83
CV	-0.12	-0.17	-0.08	-0.16	-0.13	-0.23	-0.18	-0.33
NEL	73.25	33.12	145.48	40.95	56.88	18.28	31.22	9.45

Statistics	83 617/622							
	De		Fo		NF		Wa	
	VH	VV	VH	VV	VH	VV	VH	VV
Min	-35.95	-33.92	-30.51	-25.09	-29.48	-25.75	-31.13	-27.32
Max	9.63	26.75	-0.01	12.04	7.15	17.13	12.48	29.73
Mean	-15.75	-9.87	-14.07	-7.92	-15.91	-10.02	-23.08	-19.34
SD	1.82	1.82	1.13	1.17	1.94	2.03	3.28	4.26
CV	-0.12	-0.18	-0.08	-0.15	-0.12	-0.20	-0.14	-0.22
NEL	75.23	29.50	154.10	45.80	66.95	24.39	49.46	20.64

Statistics	141 613/617							
	De		Fo		NF		Wa	
	VH	VV	VH	VV	VH	VV	VH	VV
Min	-32.53	-29.55	-29.11	-24.77	-28.65	-23.95	-30.05	-27.26
Max	2.05	13.29	0.00	6.49	0.00	15.69	0.0000	1.49
Mean	-17.26	-11.30	-14.24	-7.92	-14.02	-8.25	-19.85	-15.14
SD	2.57	2.50	1.43	1.50	2.03	2.16	4.10	5.28
CV	-0.15	-0.22	-0.10	-0.19	-0.14	-0.26	-0.21	-0.35
NEL	45.22	20.34	98.91	27.75	47.86	14.60	23.40	8.22

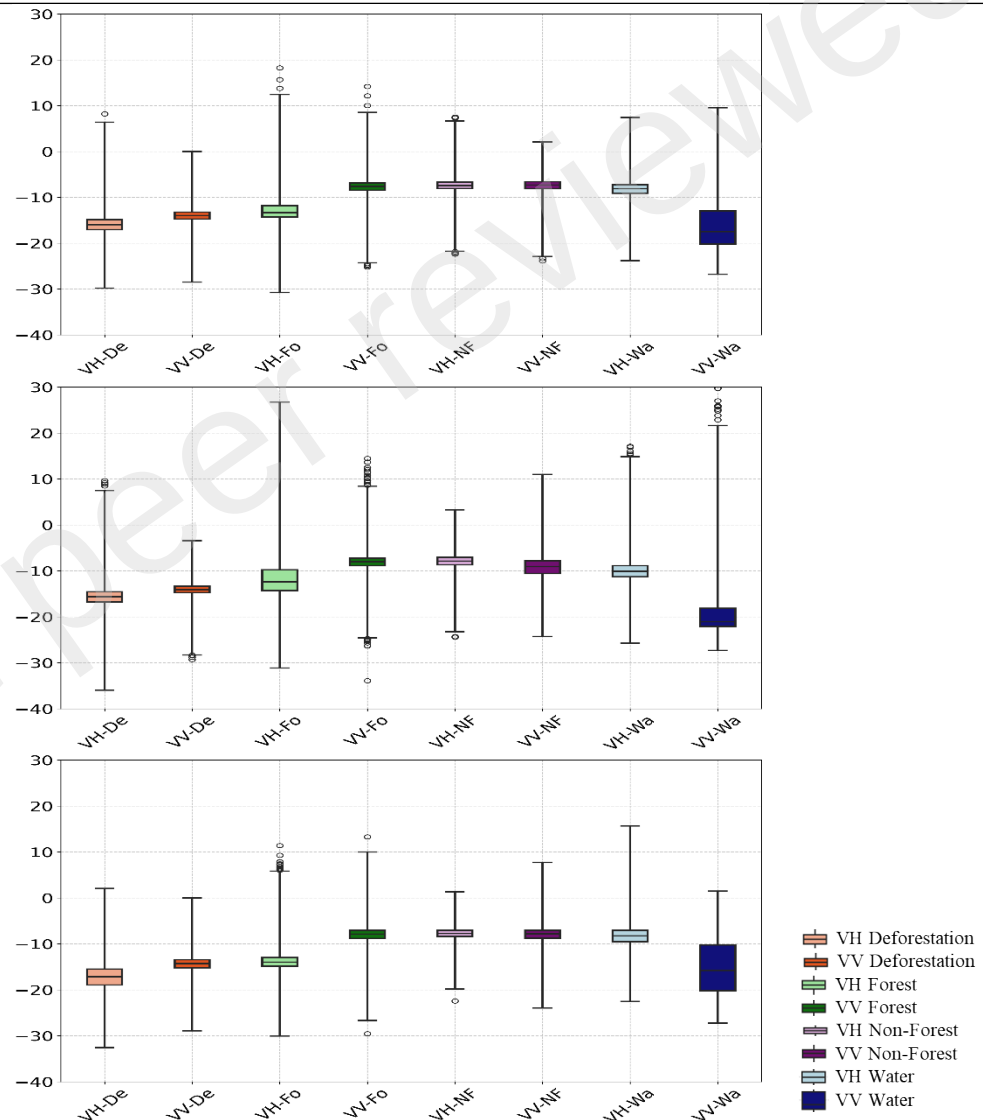


Table 7 - The Mann-Whitney-Wilcoxon and Lavene statistical tests displayed the p-values in bold that are considered statistically similar (p-value ≥ 0.05) at a 95% confidence level. The Jeffries Matusita distance has estimated the binary separability, where values below 1 are partially distinct, and values above 1 (in bold) and as close as 2 are more distinctive.

Sentinel 1		Mann-Whitney-Wilcoxon				Levene				Jeffries Matusita				
		De	Fo	NF	Wa	De	Fo	NF	Wa	De	Fo	NF	Wa	
54 622/627	VH	De	-	0.00	0.00	0.00	-	0.00	0.00	0.00	-	0.45	0.22	0.61
		Fo	0.00	-	0.26	0.00	0.00	-	0.00	0.00	0.45	-	0.10	1.13
		NF	0.00	0.26	-	0.00	0.00	0.00	-	0.00	0.22	0.10	-	0.87
		Wa	0.00	0.00	0.00	-	0.00	0.00	0.00	-	0.61	1.13	0.87	-
	VV	De	-	0.00	0.00	0.00	-	0.00	0.04	0.00	-	0.61	0.19	0.71
		Fo	0.00	-	0.00	0.00	0.00	-	0.00	0.00	0.61	-	0.16	1.18
		NF	0.00	0.00	-	0.00	0.04	0.00	-	0.00	0.19	0.16	-	0.86
		Wa	0.00	0.00	0.00	-	0.00	0.00	0.00	-	0.71	1.18	0.86	-
83 617/622	VH	De	-	0.00	0.02	0.00	-	0.00	0.07	0.00	-	0.32	0.00	0.74
		Fo	0.00	-	0.00	0.00	0.00	-	0.00	0.00	0.32	-	0.36	1.17
		NF	0.02	0.00	-	0.00	0.07	0.00	-	0.00	0.00	0.36	-	0.72
		Wa	0.00	0.00	0.00	-	0.00	0.00	0.00	-	0.74	1.17	0.72	-
	VV	De	-	0.00	0.88	0.00	-	0.00	0.02	0.00	-	0.39	0.00	0.88
		Fo	0.00	-	0.00	0.00	0.00	-	0.00	0.00	0.39	-	0.40	1.24
		NF	0.88	0.00	-	0.00	0.02	0.00	-	0.00	0.00	0.40	-	0.85
		Wa	0.00	0.00	0.00	-	0.00	0.00	0.00	-	0.88	1.24	0.85	-
141 613/617	VH	De	-	0.00	0.00	0.00	-	0.00	0.00	0.00	-	0.54	0.48	0.24
		Fo	0.00	-	0.00	0.00	0.00	-	0.00	0.00	0.54	-	0.08	0.91
		NF	0.00	0.00	-	0.00	0.00	0.00	-	0.00	0.48	0.08	-	0.80
		Wa	0.00	0.00	0.00	-	0.00	0.00	0.00	-	0.24	0.91	0.80	-
	VV	De	-	0.00	0.00	0.00	-	0.00	0.00	0.00	-	0.58	0.38	0.39
		Fo	0.00	-	0.00	0.00	0.00	-	0.00	0.00	0.58	-	0.07	0.98
		NF	0.00	0.00	-	0.00	0.00	0.00	-	0.00	0.38	0.07	-	0.78
		Wa	0.00	0.00	0.00	-	0.00	0.00	0.00	-	0.39	0.98	0.78	-

The separability distance has shown the Water to be more straightforward to classify, and also, the pair Forest/Water. These occurred in the path/frame 54 622/627 and 83 617/622. The least discernable scene was the 141 613/617. Conversely, the Non-Forest had the lowest separability, specifically in the Forest/Non-Forest pair. The other classes behaved as a partially separable target. None of the scenes have achieved higher values of separability (≥ 1.5). The multidimensional scaling approach was applied to depict the separability decomposed in two dimensions as stated by Buja et al (2008) in Figure 7.

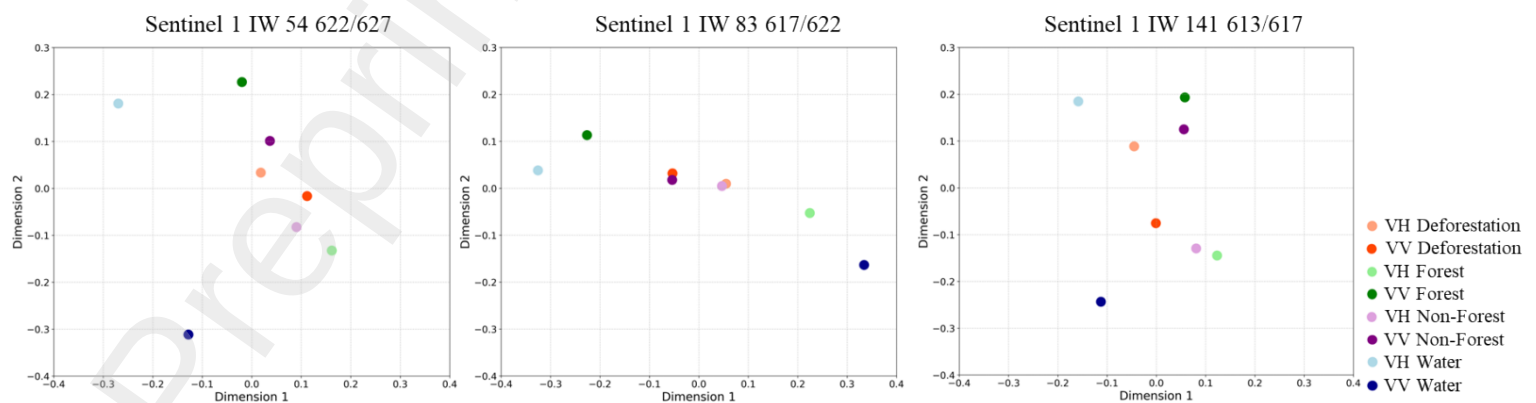


Figure 7 - Multidimensional scaling graphic to depict the Jeffries Matusita distance between the classes (Deforestation, Forest, Non-Forest and Water) in each polarization (VH, VV).

Previous studies of SAR image separability stated that the backscatter intensity has constraints to distinguish several LULC classes (Hansen et al., 2020; Liao et al., 2021). It is due to the inherent design of the SAR signal that focuses on the electric-geometric properties of targets, it is ambiguous in range and azimuth directions, it has a tradeoff between spatial resolution and swath, and, it has the speckle multiplicative noise in the resolution cell (Song et al., 2024). Sentinel 1 separability was studied also by Cremer et al. (2020), this study noticed that in backscattering time series, the VH channel is more discernable than the VV channel observing deforestation and stable forests. The improvement in recognition and separability requires the usage of time series, coherence, full polarimetric, high resolution and revisit, and flexible acquisition geometry.

The classes should behave differently regarding mean and variance for reliable LULC mapping. If the classes have a slight difference in average and high dispersion with overlap, this brings problems to the mapping. Some studies solve this problem by increasing the features, most using optical features by fusion (Reiche et al., 2018; Reis et al., 2020; Solórzano et al., 2023; Tavares et al., 2019). SAR-based systems have been overwhelming these classification issues by exploiting advanced techniques such as interferometry, polarimetry, and tomography (D'Hondt et al., 2018; Guimarães et al., 2020; Hariharan et al., 2016; Sugimoto et al., 2022).

The usage of SAR coherent features was ratified by SHAP's explanation which was observed in the three different scenes/landscapes (Figure 8). There was the highest positive contribution provided by Entropy in all path/frames, followed by Anisotropy, VV intensity, and VH coherence. Negative contributors have affected the inference like Alpha, VV coherence, and VV intensity which were oscillating positions in the features rank.

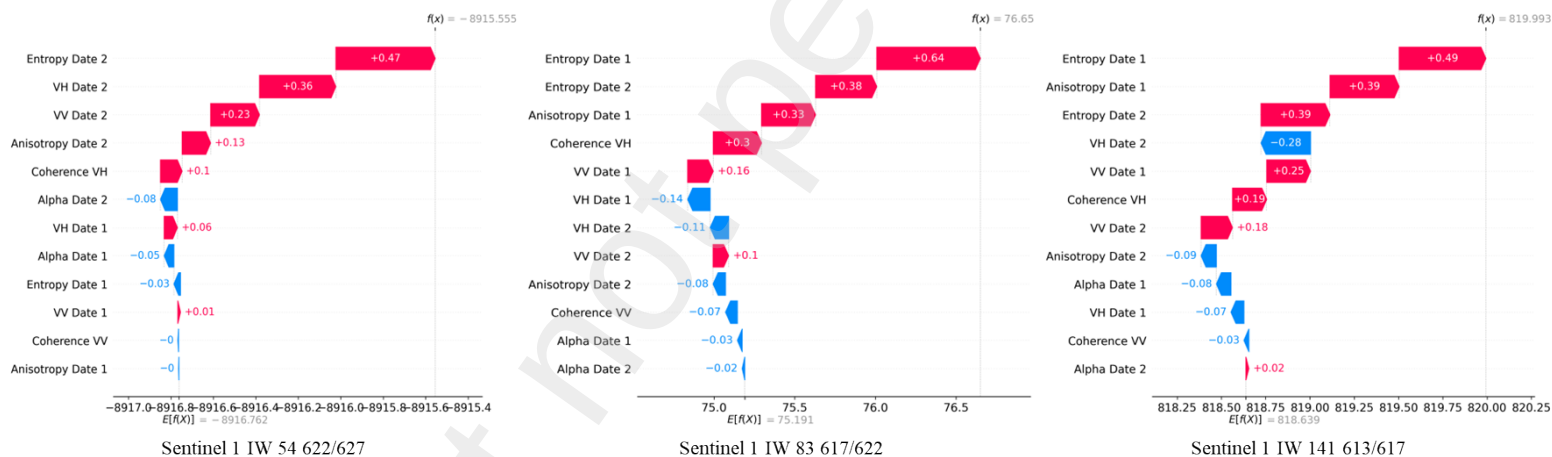


Figure 8 - Sentinel 1 Coherent features ranked by importance in the DL model prediction. The red bars displayed features that were more important to bring the model upward to the baseline, and the blue bars showed the features that moved the model downward to the baseline.

Several studies with SAR applied for deforestation in cloud computation do not consider coherent features that could have a big impact on the models (Doblas et al., 2020; Kilbride et al., 2023; Mullissa et al., 2024). The role of SAR coherent features is valuable in mapping and change detection (Guimarães et al., 2020; Jin et al., 2014; Maxwell et al., 2018). The focus on applying coherent features shall be on diminishing errors from meteorological and topographic artifacts, removing redundancies in variables, and exploiting new variables set by solving a specific scenario like water/forest or forest/non-forest.

4.2 LULC Mapping and Evaluation

Sentinel 1 coherent features based on VH and VV backscattering, interferometric coherence, and Cloude-Pottier polarimetric decomposition have allowed achieving accuracies

between [0.61, 0.95] for LULC mapping on the southern border of Amazon, under MMU of 0.0030 ha, different landscapes, and during the dry season (Table 8). The agreement metrics (global accuracy and kappa) from the confusion matrix were estimated against the optical high-resolution images from PlanetScope, and the error metrics were used to obtain the magnitude and dispersion of the error.

Table 8 - The evaluation was carried out in 18 DL models, regarding the technique (U-Net, SF-Net, and LSTM), the number of LULC classes (3 or 4), and the 3 different Sentinel 1 path/frames. The metrics of agreement and error were used, all of them obtained by the confusion matrix. The values in bold are the highest in each path/frame.

Evaluation Metrics	Classes	54 622/627			83 617/622			141 613/617		
		U-Net	SF-Net	LSTM	U-Net	SF-Net	LSTM	U-Net	SF-Net	LSTM
Accuracy	3	0.91	0.89	0.74	0.83	0.79	0.68	0.93	0.86	0.67
	4	0.95	0.88	0.73	0.69	0.73	0.61	0.88	0.74	0.66
Kappa	3	0.76	0.71	0.44	0.67	0.58	0.39	0.67	0.55	0.26
	4	0.87	0.71	0.44	0.41	0.50	0.30	0.36	0.38	0.27
Mean Error	3	0.07	13.30	44.29	-0.08	23.45	42.77	0.05	23.71	71.95
SD Error	4	0.01	14.85	44.81	0.18	35.68	50.43	0.08	25.81	38.59
SD Error	3	0.30	56.53	96.47	0.48	73.46	95.06	0.28	73.96	114.69
RMSE	4	0.34	59.54	96.89	0.78	88.19	101.22	0.38	76.58	91.05
RMSE	3	0.31	0.38	0.56	0.48	0.50	0.60	0.29	0.42	0.61
	4	0.34	0.43	0.60	0.80	0.70	0.83	0.39	0.63	0.72

The 18 DL models were elaborated considering the architecture such as U-Net, SF-Net, and LSTM, the number of classes (3 or 4), and the Sentinel 1 path/frame. The best performances were obtained by the U-Net models with accuracies of [0.69, 0.95], and the less precise models were observed in the LSTM with RMSE of [0.56, 0.83]. Regarding the Sentinel 1 path/frame, the classifiers mapped straightforwardly in scene 54 622/627 with fewer composition of Non-Forest areas, and it was harder to map scene 83 617/622 with the largest area of Non-Forest.

The mean and standard deviation of the errors have shown values between [-0.08, 71.95] and [0.28, 114.69] respectively. It is noticed that the presence of systematic error has overtaken the random errors, given by the higher values encountered in the SF-Net and LSTM models. This brings the necessity to increase the training dataset in time-space (scenes and dates) and provide more interactions (patch size and epochs) in the training loop for the more complex DL models such as SF-Net and LSTM.

Despite the accuracy being satisfactory [0.61, 0.95], it was very significant to look at the errors to reconfirm the importance of SAR geometry definition to balance the sources of error that come up with the topography, landscape, and seasonality as stated by Guimarães et al. (2020) and Koyama et al. (2019). In the perspective of increasing the accuracy and precision, employ shallower incidence angles ($>40^\circ$), mask non-forest landscapes, higher spatial resolution (3 to 6 m), update the topographic model (yearly), and obtain samples during the rainy and dry seasons.

The DL models' majority consisted of 3 classes (Deforestation, Forest, and Water), except by the model U-Net 4C in the path/frame 54 622/627 with 4 classes (Deforestation, Forest, Non-Forest, and Water) (Figure 9). This suggests that the DL models have encountered difficulties in mapping the Non-Forest class. Regarding the visual aspect of the LULC mapping, the level of noise in the LULC maps increased significantly in the LSTM models. The Water class was poorly mapped in scene 83 617/622 by the U-Net 3C model. Deforestation was mapped incorrectly in some areas with high heights and steeper topography and most of them are located in scene 141 613/617.



Figure 9 - The most accurate LULC maps encountered on each Sentinel 1 path/frames, DL models, and different numbers of classes.

DL algorithms yielded highly precise LULC maps in tropical areas, with an overall accuracy of around [0.97, 0.99] (Dalagnol et al., 2023; Doblas et al., 2020; Mullissa et al., 2023; Wagner et al., 2023). That is an impressive achievement for deforestation mapping, nevertheless, the mapping was simplified where the datasets do not cover different sorts of classes (2 and 3), landscapes (flat relief or sub-tropical areas), usage of masks, and mapping during the dry season. These previous studies that support online platforms of LULC monitoring should have handed in the polygons, promoting cross-validation.

The IoU was the most important evaluation metric considered in the development of the DL models, especially as a criterion of optimization in the training phase following DL studies of multi-class segmentation (Ramos and Sappa, 2025; Tadesse et al., 2024; Wang et al., 2022).

The evaluation of DL models handed in promising IoU values, which it was possible to consider the average by the classes such as Deforestation, Forest, Non-Forest, and Water, respectively in order of 0.54, 0.76, 0.11, and 0.22 (Figure 10). In addition, the average of IoU was checked by the models U-Net, SF-Net, and LSTM, respectively in order of 0.51, 0.45, and 0.34. The highest classification precision was found for the Forest, it takes into consideration that it is more homogenous and stable when compared to the other mapped classes. Deforestation had a satisfactory performance with IoU at [0.27, 0.84], nevertheless, it can be improved by increasing the samples or decomposing in other classes like urban areas, bare soil, and secondary vegetation. Non-Forest was the major source of errors in the classification, it can be masked or decomposed for other classes like grassland, shrub forests, swamp, and sand bars. Water appeared difficult to map, and it was related to the low level of water bodies in the Amazon during the dry season.

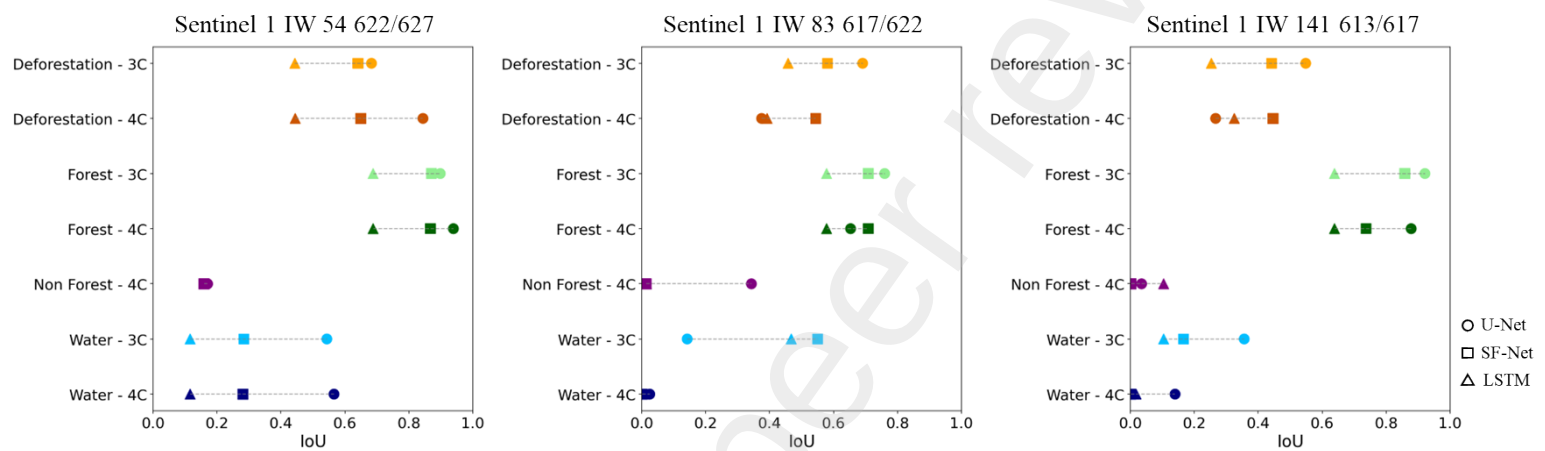


Figure 10 - The most accurate LULC maps are based on the Sentinel 1 path/frame, DL models, and different numbers of classes.

The major sources of error were noticed in the Water and Non-Forest classes, this is similar to previous studies conducted in X-band and L-band which joined the classes in the same class and used coherence properties and an updated DSM to improve the classification (Martone et al., 2018; Shimada et al., 2014). It is clear the need for extra side features capable of distinguishing firstly the forest class and after that non-forest using time-series (>5 dates), coherence (≥ 6 days), DSM (high-resolution), and longer wavelengths (L and P-bands).

The IoU has been very appropriate to quantify the performance of the LULC segmentation emphasizing the weakness and strength of the algorithm based on agreement and shape of features (Lee et al., 2020; Tadesse et al., 2024). This research found low values for IoU, which suggests the need to have extra effort in the next initiatives for training samples and stricter DL parameters for a single model which be focused on a robust model with higher performance, scale, and generalization. It is recommended to widen the usage of IoU for the assessment of multi-class segmentation models, addressing the discussion in mapping and change detection, allowing comparison, and also improving reliability and reproducibility.

The metrics of F-1 score, precision, and recall were imprinted per scene and LULC class in Figure 11. In scene 54 622/627 and the U-Net model achieved the higher values of F-1 score with [0.20, 0.97] and [0.05, 0.97], respectively. The highest precision and recall were found for the Forest in order of [0.71, 0.97] and [0.66, 0.99], respectively. It leads to the consideration that it had straightforward machine learning from the SAR features, and it is the most homogenous and stable class when compared to the other mapped classes. Deforestation has behaved at a moderate precision, and recall with [0.28, 0.98] and [0.27, 0.91], respectively. Water and Non-Forest classes have achieved a weak to moderate precision and recall in order of [0.01, 0.75] and [0.01, 0.83], respectively. It was noticed for these classes with weak to

moderate performance, the need for the extra effort in acquiring more samples for the machine learning process, and specifically, an improvement in the training modeling regarding the usage of stricter parameters to avoid the false negatives and overfitting.

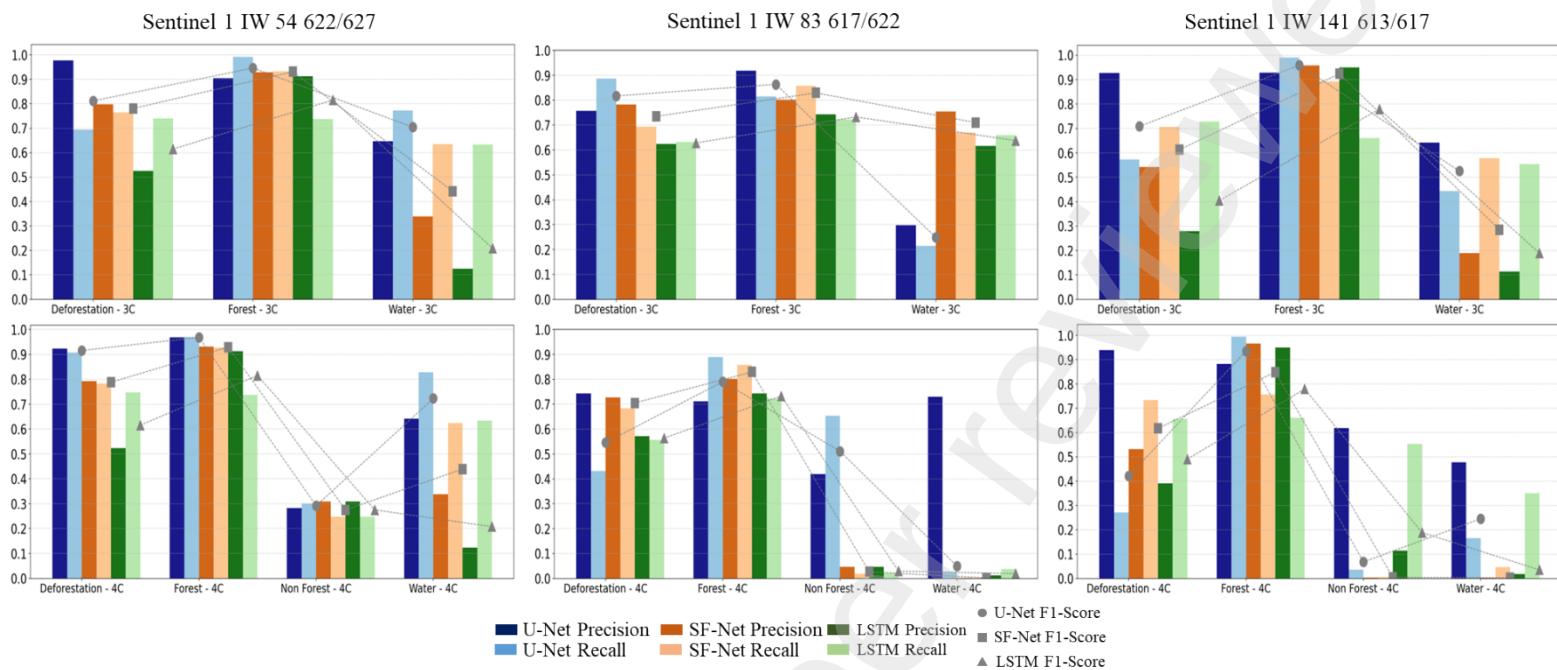


Figure 11 - The metrics of the agreement were handed in by F-1 Score, Precision, and Recall, arranged by Sentinel 1 path/frame and LULC classes.

The systematic error has overtaken the random errors considering the SF-Net and LSTM models in this research, and many studies applied to LULC mapping do not show all errors encountered. However, it is possible to discuss some aspects by the comparison between convolutional and recurrent neural networks in which simpler models can converge learning first the complex models (Wang et al., 2022, 2016). As stated by (Dal Molin and Rizzoli, 2022) the gain in performance may be reached by the attempts to find the proper parameters given by a multiscale approach based on different patches, epochs, batch sizes, and time span of the imagery.

The Taylor's and Target diagrams dialed with broad issues to rank the models regarding dispersion, matching, bias, and the magnitude of the error, specifically for the models without systematic error, which in this research were the U-Net DL models (Figure 12). As shown in Figure 12 (a), from all U-Net models, the 83 617/622 U-Net 3C achieved the shortest distance towards the GCP, considering the normalized standard deviation (σ^*) in order of 1.73 and the moderate correlation coefficient (ρ) in order of 0.43. This means that the U-Net model learned straightforward the variability of scene 83 617/622, implying a precise inference in the shape of the features in terms of repeated measurements, but not into the true value. On the other hand, the correlation in 83 617/622 was not stronger than in models in scene 54 622/627, the dispersion has more impact on the model performance than linear association and it is explained by the difference of area in the scenes by Non-Forest class. The largest σ^* was between [2.48, 3.28] for the scene 141 613/617 which yielded the U-Net DL models further away from the reference (GCP) gathering less precision, despite good values of ρ between [0.38, 0.65], this harder learning of the U-Net models in this scene are related the topography more undulated and hillier.

Taking into account the Target diagram in Figure 12 (b), the majority of U-Net models overestimated the inference of LULC classes presenting on B^* of $[0.01, 0.18]$, except by the 83 617/622 U-Net 3C that yielded an underestimated modeling with B^* of -0.08. The models 83 617/622 U-Net 4C and 141 613/617 U-Net 4C have reached higher magnitudes of error that placed them further away from the GCP with $uRMSE^*$ values around 0.78. The best model positions were allocated by 54 622/627 U-Net 4C, 54 622/627 U-Net 3C, and 83 617/622 3C, this is related to the minor or removal of areas of Non-Forest and their relief characterized as flat to undulated.

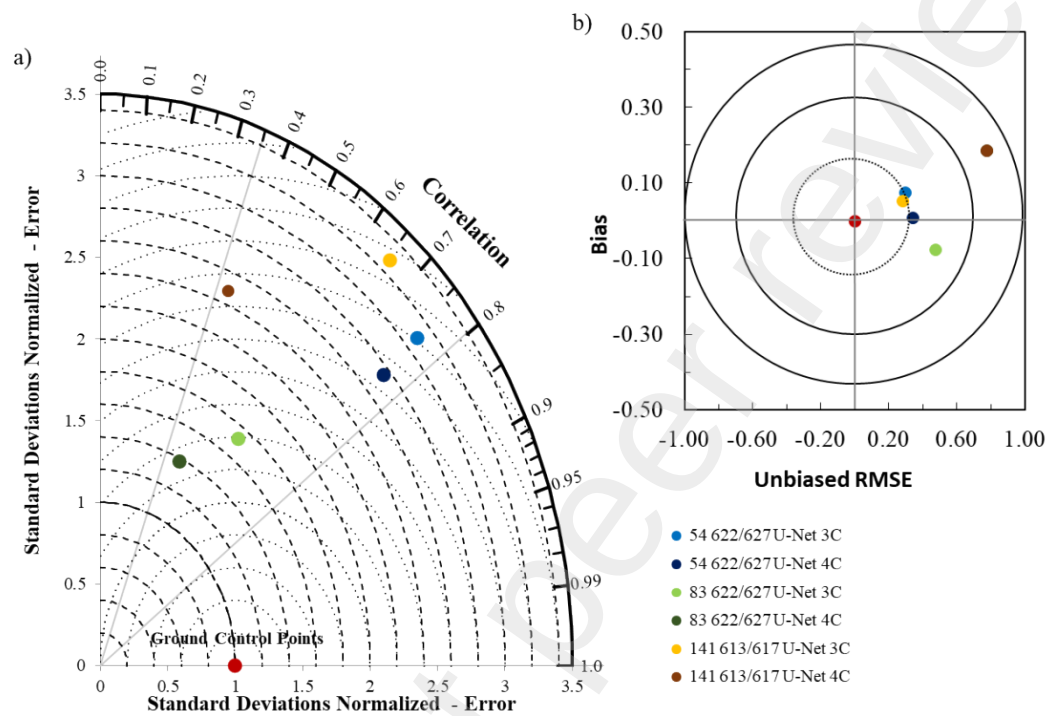


Figure 12 - The Taylor (a) and the Target (b) diagrams for Sentinel 1 scenes and U-Net models revealed the position per correlation and error regarding the GCP observed in optical images.

In general, the 54 622/627 U-Net 4C model alternated the lower and higher positions on Taylor and Target diagrams. However, it accomplished the best ranking if compared to the other scenes and models. Conversely, it appears that 141 613/617 U-Net 4C yielded the weakest correlation and highest values of error. Several DL studies pointed out that it must prioritize the variability of the dataset and after that, there are some techniques focused on improving borders, contours, and lines such as conditional random field, multi-scale, multi-patches, pixel-wise segmentation, and the promising complex numbers for DL (Huang et al., 2020; Liu et al., 2023; Ren et al., 2023; Wang et al., 2021).

Therefore, if the goal is to move ρ and σ^* forward to the GCP and to decrease the impact from bias and $uRMSE^*$, then robust learning is needed based on stricter hyperparameters after longer training loops and precise acquisition geometry fitted to temporal and spatial sort of landscapes considering shallow incidences angles and multiple SAR passes.

This research has brought to discussion the significance of the EO system, the landscape, and the algorithms developed to map deforestation and/or LULC, which depicts the trends for remote sensing, techniques, and performances (it may be overall accuracy, precision, recall and coefficient of determination) (Table 10). Specifically, it has highlighted Sentinel 1 and its coherent features have accomplished performance among fair to excellent, using training and inference of DL modeling for a sample of Amazon with a high diversity of landscapes. It is noticed that previous studies can be organized into two groups: i) optical-based EO systems

using high-resolution and DL models (Dalagnol et al., 2023; Hansen et al., 2016; Maretto et al., 2021; Wagner et al., 2023); and ii) radar-based EO systems using probabilistic and Bayesian techniques (Ballère et al., 2021; Doblas et al., 2020; Mermoz et al., 2021; Watanabe et al., 2018), except by Mullissa et al. (2024) which has been using LSTM model. There are available precise and high spatial resolutions LULC mappings based on optical systems, however, they are not validated in areas between -4 up to 4 degrees.

It is interesting to note in Table 10 that several studies mapped large areas of coverage (> 1 million km²), and high-performance levels (>0.83) were obtained. The higher revisit time is given by radar-based EO systems between 6 up to 12 days, and the majority of MMUs are equal to or larger than 0.1 ha, except by Mullissa et al. (2024). This research elaborated a robust DL method and reached a thematic performance equivalent to the current most precise radar-based algorithms to map deforestation as noticed in the studies from Doblas et al. (2020), Hoekman et al. (2020), Mermoz et al. (2021), Mullissa et al. (2024); Reiche et al., 2021). There is potential to achieve a reliable change detection approach, which might employ simpler techniques like subtraction and radar change ratio. Furthermore, the SLC signal allows exploiting new SAR techniques such as time series, tomography, and polarimetric-interferometry (Figure 13).

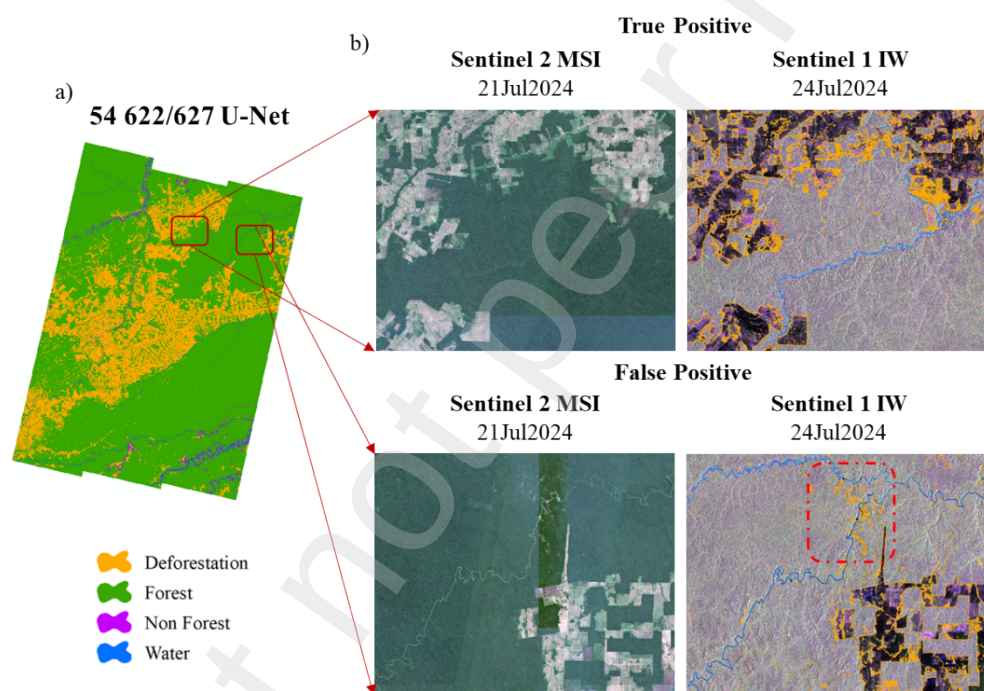


Figure 13 - The map of deforestation yielded through Sentinel 1 (a) shows areas of clear cutting (orange). The right upper part (b) depicts the true positive features checked in Sentinel 2 and Sentinel 1, and the bottom part is a false positive (dashed red).

In most previous studies in Table 9, the validation datasets do not have better error conditions. Cross-validation uses previous mapping, in which the error is not well-known, nor synchronized in the time spans and without better spatial resolution. In these cases, the error is propagated between the datasets, specifically in areas with high geometric and radiometric error and outdated DEM. It ratifies the significance of developing more ML/DL algorithms regarding the new upcoming SAR systems (Sentinel 1C, Nisar, Alos-4, Biomass), which may support alternative SAR tools by coherence features to estimate forest disturbance, regrowth, biomass, and updated DSM (Guimarães et al., 2018; Soja et al., 2021b, 2018).

Table 9 - The studies focused on mapping deforestation and LULC by applying optical and SAR systems which were assessed in tropical landscapes.

Author	Environment	EO System	Features	Classifier	Classes	Revisit	Coverage (km ²)	MMU (ha)	Accuracy
This Research	Tropical Forest Flat, Undulated and Hilly Reliefs	Sentinel 1, SAR C-Band	VV+VH, Coherence, Polarimetry	U-Net, SF-Net, LSTM	4	12	246.38 x 10 ³	0.0030	0.61-0.95
Mullissa et al. (2024)	Tropical, Subtropical and Boreal Forests Flat, Undulated, Hilly and Mountainous Reliefs	Sentinel 1, SAR C-Band	VV+VH, Time-Series	LSTM, Probability Detection	2	12	149.0 x 10 ⁶	0.0500	0.63-0.83
Dalagnol et al. (2023)	Tropical Forest Flat and Undulated Reliefs	PlanetScope MSI VNIR	1-4 Bands, Reflectance	U-Net	3	30	507.0 x 10 ³	0.0005	0.69-0.97
Wagner et al. (2023)	Tropical Forest Flat and Undulated Reliefs	PlanetScope MSI VNIR	1-4 Bands, Reflectance	U-Net	2	180	1.0 x 10 ⁶	0.0005	0.67-0.98
Ballère et al. (2021)	Tropical Forest Flat, Undulated Reliefs	Sentinel 1, SAR C-Band	VV+VH, Time-Series	Radar Change Ratio	2	6	83.5 x 10 ³	0.1000	0.82-0.96
Maretto et al. (2021)	Tropical, Flat, Undulated Reliefs	Landsat 8, OLI VNIR, SWIR	1-9 Bands, Reflectance	U-Net	2	16	133.2 x 10 ³	0.0030	0.94-0.95
Mermoz et al. (2021)	Tropical Forest, Savanah and Grassland Flat, Undulated, Hilly and Mountainous Reliefs	Sentinel 1, SAR C-Band	VV+VH, Time-Series	Radar Change Ratio	2	6	750.0 x 10 ³	0.1000	0.95-0.99
Reiche et al. (2021)	Tropical Forest Flat, Undulated and Hilly Reliefs	Sentinel 1, SAR C-Band	VV+VH, Time-Series	Gaussian Mixture Models, Bayesian Technique	2	6	1.9 x 10 ⁶	0.2000	0.84-0.98
Watanabe et al. (2021)	Tropical Forest, Savanah and Grassland Flat, Undulated, Hilly and Mountainous Reliefs	Alos 2, SAR L-Band	HH+HV, Time-Series	Change Detection Threshold	3	42	24.0 x 10 ⁶	2.0000	0.64-0.84
Doblas et al. (2020)	Tropical Forest Flat, Undulated	Sentinel 1, SAR C-Band	HH+HV, Time-Series	Adaptive Linear Threshold	2	12	5.2 x 10 ⁶	0.1000	0.94-0.96
Hoekman et al. (2020)	Tropical Forest Flat, Undulated	Sentinel 1, SAR C-Band	HH+HV, Time-Series	Bayesian Technique	4	6	135.0 x 10 ³	0.3000	0.86-0.99
Hansen et al. (2016)	Tropical Forest, Savanah and Grassland Flat, Undulated, Hilly and Mountainous Reliefs	Landsat 7/8, ETM, OLI VNIR, SWIR	1-9 Bands, Reflectance	Decision Tree	2	16	4.3 x 10 ⁶	0.0030	0.67-0.99
Diniz et al., (2015)	Tropical Forest Flat, Undulated	IRS-P6, AWFIS, VNIR, SWIR	1-4 Bands, Reflectance	Visual Interpretation	8	5	5.2 x 10 ⁶	0.3136	0.83

5. Conclusions

The Sentinel 1 C-band SAR dataset provided the LULC mapping of the southern Amazon, and it achieved satisfactory reliability at three different levels of landscape (path/frames), under high deforestation pressure. 18 DL models were elaborated using the backscattering and coherent features (interferometric coherence and dual-polarization decomposition), small temporal decorrelation (12 days), and moderate incidence angle (35°), besides the moderate spatial resolution (30 m). It was accepted the assumption that the DL models had the thematic error lying in the location of the scenes and the kind of LULC class, during the dry season. The highest mapping accuracy (0.95) and IoU (0.66) was reached by Sentinel 1 scene 54 622/627, U-Net model, and 4 classes, and the lowest mapping accuracy (0.61) and IoU (0.36) was noticed by Sentinel 1 scene 83 617/622, LSTM model and 4 classes.

The focus of this research was elaborating a robust DL method for Sentinel 1 rather than obtaining robust models, although the performances were equivalent to studies on the same theme. The limitation of DL models relies on the employment of the inference in different Sentinel 1 scenes on Amazon that are out of the study area. The DL model shall be improved to achieve better scales, generalization, and performance by increasing spatial (scenes) and time (acquisition dates) context.

The LULC classes were described by backscattering coefficient which shared a lot of ambiguity and combined with high relative dispersion, as in the worst case noticed in the Non-Forest class, which made the mapping difficult to obtain separability and high thematic precision (omissions and commissions errors). The Forest and Deforestation were more discernable LULC classes, and the flat to undulated relief corroborated with their mapping.

It is important to emphasize five aspects: i) Sentinel 1 path/frame have behaved differently regarding the thematic error where the precision is higher in flat to undulated topography without non-forest areas like savannahs, shrub forests and/or grassland; ii) The interface between land and water are difficult to map in the occasion of low levels of water (droughts), and it shall be considered a mask; iii) increase the samples in the training modeling using more diversity in the landscape and seasonality (dry and rainy); iv) the U-Net model have caught the variability of datasets straightforward in the training modeling trough the low values set in batch size, patch size and epochs, conversely the SF-Net and LSTM models had strong ambiguity to classify LULC using the same hyperparameters, which suggests more tests for the complex models and giving more time to obtain convergence and avoid overfitting; v) nonconformities from LULC mapping shall be minimized by the usage of shallow incidence angles ($>45^\circ$), orbital passing in descending and ascending, usage of updated DSM (yearly) and finer spatial resolution (3 to 6 m), radar vegetation index, texture in dual channels, and longer time series (scenes >15). The main effort in this study was to elaborate training and inference models that apply Sentinel 1 in LULC mapping.

New initiatives in the Amazon shall consider the newer SAR systems with longer wavelengths (S, L, and P), better temporal decorrelation (less than 12 days), and full polarimetric channels (quad-pol), and firstly to improve the mapping focused on the steady tropical forest. It is possible to achieve a detailed level to detect deforestation under a high diversity of landscapes through coherent features. It is expected to increase the SAR dataset samples providing a longer time sequence, and this will increase the thematic performance for generalization, scalability, and mapping the forest disturbance in the Amazon.

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References

- Ab'Saber, A.N., 2002. Bases para o estudo dos ecossistemas da Amazônia brasileira. *Estudos Avançados* 16, 7–30. <https://doi.org/10.1590/S0103-40142002000200002>
- Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: A Next-generation Hyperparameter Optimization Framework.
- Altarex, R.D.D., Apan, A., Maraseni, T., 2023. Deep learning U-Net classification of Sentinel-1 and 2 fusions effectively demarcates tropical montane forest's deforestation. *Remote Sens Appl* 29, 100887. <https://doi.org/10.1016/j.rsase.2022.100887>
- Ballère, M., Bouvet, A., Mermoz, S., Le Toan, T., Koleček, T., Bedeau, C., André, M., Forestier, E., Frison, P.-L., Lardeux, C., 2021. SAR data for tropical forest disturbance alerts in French Guiana: Benefit over optical imagery. *Remote Sens Environ* 252, 112159. <https://doi.org/10.1016/j.rse.2020.112159>
- Barreto, P., Pereira, R., Arima, E., 2008. A pecuária e o desmatamento na Amazônia na era das mudanças climáticas. Instituto do Homem e Meio Ambiente da Amazônia.
- Belgiu, M., Drăgu, L., 2016. Random forest in remote sensing: A review of applications and future directions. *ISPRS Journal of Photogrammetry and Remote Sensing*. <https://doi.org/10.1016/j.isprsjprs.2016.01.011>
- Breiman, L., 2001. Random Forests. *Mach Learn* 45, 5–32.
- Canisius, F., Brisco, B., Murnaghan, K., Van Der Kooij, M., Keizer, E., 2019. SAR Backscatter and InSAR Coherence for Monitoring Wetland Extent, Flood Pulse and Vegetation: A Study of the Amazon Lowland. *Remote Sens (Basel)* 11, 720. <https://doi.org/10.3390/rs11060720>
- Chatziantoniou, A., Psomiadis, E., Petropoulos, G., 2017. Co-Orbital Sentinel 1 and 2 for LULC Mapping with Emphasis on Wetlands in a Mediterranean Setting Based on Machine Learning. *Remote Sens (Basel)* 9, 1259. <https://doi.org/10.3390/rs9121259>
- Chaves, M.E.D., Mataveli, G., Conceição, K. V., Adami, M., Petrone, F.G., Sanches, I.D., 2024. AMACRO: the newer Amazonia deforestation hotspot and a potential setback for Brazilian agriculture. *Perspect Ecol Conserv* 22, 93–100. <https://doi.org/10.1016/j.pecon.2024.01.009>
- Clark, A., Phinn, S., Scarth, P., 2023. Optimised U-Net for Land Use–Land Cover Classification Using Aerial Photography. *PFG - Journal of Photogrammetry, Remote Sensing and Geoinformation Science* 91, 125–147. <https://doi.org/10.1007/s41064-023-00233-3>
- Congalton, R.G., Green, K., 2019. Assessing the Accuracy of Remotely Sensed Data. CRC Press, New York. <https://doi.org/10.1201/9780429052729>
- Cremer, F., Urbazaev, M., Cortes, J., Truckenbrodt, J., Schmullius, C., Thiel, C., 2020. Potential of Recurrence Metrics from Sentinel-1 Time Series for Deforestation Mapping. *IEEE J Sel Top Appl Earth Obs Remote Sens* 13, 5233–5240. <https://doi.org/10.1109/JSTARS.2020.3019333>
- CTREE, 2024. REDD+AI [WWW Document]. REDD+AI.
- Dal Molin, R., Rizzoli, P., 2022. Potential of Convolutional Neural Networks for Forest Mapping Using Sentinel-1 Interferometric Short Time Series. *Remote Sens (Basel)* 14. <https://doi.org/10.3390/rs14061381>

- Dalagnol, R., Wagner, F.H., Galvão, L.S., Braga, D., Osborn, F., Sagang, L.B., da Conceição Bispo, P., Payne, M., Silva Junior, C., Favrichon, S., Silgueiro, V., Anderson, L.O., Aragão, L.E.O. e C. de, Fensholt, R., Brandt, M., Ciais, P., Saatchi, S., 2023. Mapping tropical forest degradation with deep learning and Planet NICFI data. *Remote Sens Environ* 298, 113798. <https://doi.org/10.1016/j.rse.2023.113798>
- Dasari, K., Anjaneyulu, L., Nadimikeri, J., 2022. Application of C-band sentinel-1A SAR data as proxies for detecting oil spills of Chennai, East Coast of India. *Mar Pollut Bull* 174, 113182. <https://doi.org/10.1016/j.marpolbul.2021.113182>
- D'Hondt, O., Hänsch, R., Wagener, N., Hellwich, O., 2018. Exploiting SAR Tomography for Supervised Land-Cover Classification. *Remote Sens (Basel)* 10, 1742. <https://doi.org/10.3390/rs10111742>
- Diaz, G.I., Fokoue-Nkoutche, A., Nannicini, G., Samulowitz, H., 2017. An effective algorithm for hyperparameter optimization of neural networks. *IBM J Res Dev* 61, 9:1-9:11. <https://doi.org/10.1147/JRD.2017.2709578>
- Diniz, C.G., Souza, A.A.D.A., Santos, D.C., Dias, M.C., Luz, N.C. Da, Moraes, D.R.V. De, Maia, J.S.A., Gomes, A.R., Narvaes, I.D.S., Valeriano, D.M., Maurano, L.E.P., Adami, M., 2015. DETER-B: The New Amazon Near Real-Time Deforestation Detection System. *IEEE J Sel Top Appl Earth Obs Remote Sens* 8, 3619–3628. <https://doi.org/10.1109/JSTARS.2015.2437075>
- Doblas, J., Shimabukuro, Y., Sant'Anna, S., Carneiro, A., Aragão, L., Almeida, C., 2020. Optimizing Near Real-Time Detection of Deforestation on Tropical Rainforests Using Sentinel-1 Data. *Remote Sens (Basel)* 12, 3922. <https://doi.org/10.3390/rs12233922>
- Dong, X., Zhang, D., Cui, K., Hu, C., Lv, X., 2016. Spatial filtering strategies on deforestation detection using SAR image textures, in: 2016 CIE International Conference on Radar (RADAR). IEEE, pp. 1–4. <https://doi.org/10.1109/RADAR.2016.8059472>
- ESA, (European Space Agency), 2024a. Sentinels Scientific Data Hub [WWW Document]. European Space Agency. URL <https://scihub.copernicus.eu> (accessed 12.14.24).
- ESA, (European Space Agency), 2024b. Sentinel's Application Platform - SNAP.
- Estimates, P., 2021. Polarimetric Synthetic Aperture Radar, Remote Sensing and Digital Image Processing. Springer International Publishing, Cham. <https://doi.org/10.1007/978-3-030-56504-6>
- FAO, 2024. The State of the World's Forests 2024, The State of the World's Forests 2024. FAO. <https://doi.org/10.4060/cd1211en>
- FAO, 2020. Global Forest Resources Assessment 2020, Second. ed, FAO Forestry. FAO. <https://doi.org/10.4060/ca8753en>
- FAO, UNEP, 2020. The State of the World's Forests 2020. FAO and UNEP, Rome. <https://doi.org/10.4060/ca8642en>
- Ferretti, A., Monti-Guarnieri, A., Prati, C., Rocca, F., Massonet, D., 2007. InSAR Principles-Guidelines for SAR Interferometry Processing and Interpretation. ESA Publications, Noordwijk.
- Fu, B., Wang, Y., Campbell, A., Li, Y., Zhang, B., Yin, S., Xing, Z., Jin, X., 2017. Comparison of object-based and pixel-based Random Forest algorithm for wetland vegetation mapping using high spatial resolution GF-1 and SAR data. *Ecol Indic* 73, 105–117. <https://doi.org/10.1016/j.ecolind.2016.09.029>
- Garcia-Garcia, A., Orts-Escolano, S., Oprea, S., Villena-Martinez, V., Martinez-Gonzalez, P., Garcia-Rodriguez, J., 2018. A survey on deep learning techniques for image and video semantic segmentation. *Applied Soft Computing Journal*. <https://doi.org/10.1016/j.asoc.2018.05.018>
- Greff, K., Srivastava, R.K., Koutník, J., Steunebrink, B.R., Schmidhuber, J., 2015. LSTM: A Search Space Odyssey. *IEEE Trans Neural Netw Learn Syst* 28, 2222–2232. <https://doi.org/10.1109/TNNLS.2016.2582924>
- Guimarães, U.S., Narvaes, I. da S., Galo, M. de L.B.T., da Silva, A. de Q., Camargo, P. de O., 2018. Radargrammetric approaches to the flat relief of the amazon coast using COSMO-SkyMed and TerraSAR-X datasets. *ISPRS Journal of Photogrammetry and Remote Sensing* 145, 284–296. <https://doi.org/10.1016/j.isprsjprs.2018.09.001>

- Guimarães, U.S., Narvaes, I.D.S., Galo, M. de L.B.T., 2017. APLICAÇÃO DE DADOS ERS, ENVISAT E SENTINEL PARA DETECÇÃO DE MUDANÇAS NOS AMBIENTES COSTEIROS AMAZÔNICOS. *Revista Brasileira de Geomorfologia* 18. <https://doi.org/10.20502/rbg.v18i2.998>
- Guimarães, U.Si., de Lourdes Bueno Trindade Galo, M., da Silva Narvaes, I., de Queiroz da Silva, A., 2020. Cosmo-SkyMed and TerraSAR-X datasets for geomorphological mapping in the eastern of Marajó Island, Amazon coast. *Geomorphology* 350, 106934. <https://doi.org/10.1016/j.geomorph.2019.106934>
- Hansen, J.N., Mitchard, E.T.A., King, S., 2020. Assessing Forest/Non-Forest Separability Using Sentinel-1 C-Band Synthetic Aperture Radar. *Remote Sens (Basel)* 12, 1899. <https://doi.org/10.3390/rs12111899>
- Hansen, M.C., Krylov, A., Tyukavina, A., Potapov, P. V., Turubanova, S., Zutta, B., Ifo, S., Margono, B., Stolle, F., Moore, R., 2016. Humid tropical forest disturbance alerts using Landsat data. *Environmental Research Letters* 11, 034008. <https://doi.org/10.1088/1748-9326/11/3/034008>
- Hansen, M.C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S.A., Tyukavina, A., Thau, D., Stehman, S. V., Goetz, S.J., Loveland, T.R., Kommareddy, A., Egorov, A., Chini, L., Justice, C.O., Townshend, J.R.G., 2013a. High-Resolution Global Maps of 21st-Century Forest Cover Change. *Science (1979)* 342, 850–853. <https://doi.org/10.1126/science.1244693>
- Hansen, M.C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S.A., Tyukavina, A., Thau, D., Stehman, S. V., Goetz, S.J., Loveland, T.R., Kommareddy, A., Egorov, A., Chini, L., Justice, C.O., Townshend, J.R.G., 2013b. High-Resolution Global Maps of 21st-Century Forest Cover Change. *Science (1979)* 342, 850–853. <https://doi.org/10.1126/science.1244693>
- Hariharan, S., Tirodkar, S., Bhattacharya, A., 2016. Polarimetric SAR decomposition parameter subset selection and their optimal dynamic range evaluation for urban area classification using Random Forest. *International Journal of Applied Earth Observation and Geoinformation* 44, 144–158. <https://doi.org/10.1016/j.jag.2015.08.007>
- He, K., Zhang, X., Ren, S., Sun, J., 2015. Deep Residual Learning for Image Recognition.
- Hoekman, D., Kooij, B., Quiñones, M., Vellekoop, S., Carolita, I., Budhiman, S., Arief, R., Roswintiarti, O., 2020. Wide-Area Near-Real-Time Monitoring of Tropical Forest Degradation and Deforestation Using Sentinel-1. *Remote Sens (Basel)* 12, 3263. <https://doi.org/10.3390/rs12193263>
- Hu, J., Shen, L., Albanie, S., Sun, G., Wu, E., 2017. Squeeze-and-Excitation Networks.
- Hu, X., Li, L., Huang, J., Zeng, Y., Zhang, S., Su, Y., Hong, Y., Hong, Z., 2024. Radar vegetation indices for monitoring surface vegetation: Developments, challenges, and trends. *Science of The Total Environment* 945, 173974. <https://doi.org/10.1016/j.scitotenv.2024.173974>
- Huang, Y., Meng, M., Hou, Z., Wu, L., Guo, Z., Shen, X., Zheng, W., Li, N., 2023. Land Cover Classification of SAR Based on 1DCNN-MRF Model Using Improved Dual-Polarization Radar Vegetation Index. *Remote Sens (Basel)* 15, 3221. <https://doi.org/10.3390/rs15133221>
- Huang, Z., Datcu, M., Pan, Z., Lei, B., 2020. Deep SAR-Net: Learning objects from signals. *ISPRS Journal of Photogrammetry and Remote Sensing* 161, 179–193. <https://doi.org/10.1016/j.isprsjprs.2020.01.016>
- IBGE, 2020a. Base geomorfológica continua do Brasil - Escala 1:250.000 [WWW Document]. Instituto Brasileiro de Geografia e Estatística. URL <https://www.ibge.gov.br/en/geosciences/downloads-geosciences.html> (accessed 11.2.24).
- IBGE, 2020b. Base geológica continua do Brasil - Escala 1:250.000 [WWW Document]. Instituto Brasileiro de Geografia e Estatística. URL http://downloads.ibge.gov.br/downloads_geociencias.htm (accessed 4.24.15).
- IBGE, 2012. Manual Técnico da Vegetação Brasileira, Instituto Brasileiro de Geografia e Estatística - IBGE. <https://doi.org/ISSN 0101-4234>
- Ienco, D., Gaetano, R., Dupaquier, C., Maurel, P., 2017. Land Cover Classification via Multi-temporal Spatial Data by Recurrent Neural Networks. <https://doi.org/10.1109/LGRS.2017.2728698>

- INMET, (Instituto Nacional de Meteorologia), 2024. Banco de Dados Meteorológicos para Ensino e Pesquisa [WWW Document]. Instituto Nacional de Meteorologia. URL <https://mapas.inmet.gov.br/> (accessed 11.26.24).
- INPE, 2024. Programa de Monitoramento da Amazônia e Demais Biomas [WWW Document]. Instituto Nacional de Pesquisas Espaciais. URL <http://terrabrasilis.dpi.inpe.br/downloads/> (accessed 10.22.24).
- Jin, H., Mountrakis, G., Stehman, S. V., 2014. Assessing integration of intensity, polarimetric scattering, interferometric coherence and spatial texture metrics in PALSAR-derived land cover classification. *ISPRS Journal of Photogrammetry and Remote Sensing* 98, 70–84. <https://doi.org/10.1016/j.isprsjprs.2014.09.017>
- Joshi, N., Baumann, M., Ehammer, A., Fensholt, R., Grogan, K., Hostert, P., Jepsen, M.R., Kuemmerle, T., Meyfroidt, P., Mitchard, E.T.A., Reiche, J., Ryan, C.M., Waske, B., 2016. A review of the application of optical and radar remote sensing data fusion to land use mapping and monitoring. *Remote Sens (Basel)* 8, 1–23. <https://doi.org/10.3390/rs8010070>
- Kang, J., Ji, T., Zhang, Z., Fernandez-Beltran, R., 2023. SAR time series despeckling via nonlocal matrix decomposition in logarithm domain. *Signal Processing* 209, 109040. <https://doi.org/10.1016/j.sigpro.2023.109040>
- Kilbride, J., Poortinga, A., Bhandari, B., Thwal, N., Quyen, N., Silverman, J., Tenneson, K., Bell, D., Gregory, M., Kennedy, R., Saah, D., 2023. Near Real-Time Mapping of Tropical Forest Disturbance Using SAR and Semantic Segmentation in Google Earth Engine. *Remote Sens (Basel)* 15, 5223. <https://doi.org/10.3390/rs15215223>
- Koyama, C.N., Watanabe, M., Hayashi, M., Ogawa, T., Shimada, M., 2019. Mapping the spatial-temporal variability of tropical forests by ALOS-2 L-band SAR big data analysis. *Remote Sens Environ* 233, 111372. <https://doi.org/10.1016/j.rse.2019.111372>
- Kuck, T.N., Silva Filho, P.F.F., Sano, E.E., Bispo, P. da C., Shiguemori, E.H., Dalagnol, R., 2021. Change Detection of Selective Logging in the Brazilian Amazon Using X-Band SAR Data and Pre-Trained Convolutional Neural Networks. *Remote Sens (Basel)* 13, 4944. <https://doi.org/10.3390/rs13234944>
- Lee, J.-S., Pottier, E., 2017. *Polarimetric Radar Imaging*. CRC Press. <https://doi.org/10.1201/9781420054989>
- Lee, S.-H., Han, K.-J., Lee, K., Lee, K.-J., Oh, K.-Y., Lee, M.-J., 2020. Classification of Landscape Affected by Deforestation Using High-Resolution Remote Sensing Data and Deep-Learning Techniques. *Remote Sens (Basel)* 12, 3372. <https://doi.org/10.3390/rs12203372>
- Li, X., You, A., Zhu, Z., Zhao, H., Yang, M., Yang, K., Tong, Y., 2020. Semantic Flow for Fast and Accurate Scene Parsing.
- Li, X., Zhang, J., Yang, Y., Cheng, G., Yang, K., Tong, Y., Tao, D., 2024. Sfnet: Faster and Accurate Semantic Segmentation Via Semantic Flow. *Int J Comput Vis* 132, 466–489. <https://doi.org/10.1007/s11263-023-01875-x>
- Liao, N., Datcu, M., Zhang, Z., Guo, W., Zhao, J., Yu, W., 2021. Analyzing the Separability of SAR Classification Dataset in Open Set Conditions. *IEEE J Sel Top Appl Earth Obs Remote Sens* 14, 7895–7910. <https://doi.org/10.1109/JSTARS.2021.3100342>
- Liesenbergh, V., de Souza Filho, C.R., Gloaguen, R., 2016. Evaluating Moisture and Geometry Effects on L-Band SAR Classification Performance Over a Tropical Rain Forest Environment. *IEEE J Sel Top Appl Earth Obs Remote Sens* 9, 5357–5368. <https://doi.org/10.1109/JSTARS.2016.2617120>
- Liesenbergh, V., Gloaguen, R., 2013. Evaluating SAR polarization modes at L-band for forest classification purposes in Eastern Amazon, Brazil. *International Journal of Applied Earth Observation and Geoinformation* 21, 122–135. <https://doi.org/10.1016/j.jag.2012.08.016>
- Liu, B., Chen, S., Gao, L., 2025. Combining residual convolutional LSTM with attention mechanisms for spatiotemporal forest cover prediction. *Environmental Modelling & Software* 183, 106260. <https://doi.org/10.1016/j.envsoft.2024.106260>
- Liu, Y., Shi, S., Wang, J., Zhong, Y., 2023. Seeing Beyond the Patch: Scale-Adaptive Semantic Segmentation of High-resolution Remote Sensing Imagery based on Reinforcement Learning.

- Lundberg, S., Lee, S.-I., 2017. A Unified Approach to Interpreting Model Predictions.
- Ma, L., Li, M., Ma, X., Cheng, L., Du, P., Liu, Y., 2017. A review of supervised object-based land-cover image classification. *ISPRS Journal of Photogrammetry and Remote Sensing* 130, 277–293. <https://doi.org/10.1016/j.isprsjprs.2017.06.001>
- Ma, L., Liu, Y., Zhang, X., Ye, Y., Yin, G., Johnson, B.A., 2019. Deep learning in remote sensing applications: A meta-analysis and review. *ISPRS Journal of Photogrammetry and Remote Sensing* 152, 166–177. <https://doi.org/10.1016/j.isprsjprs.2019.04.015>
- MapBiomas, 2024. Collection 9 of the 2024 series of Maps of Land Cover and Use and Fire Scar of Brazil [WWW Document]. MapBiomas. URL <https://brasil.mapbiomas.org/en/produtos/> (accessed 11.24.24).
- Maretto, R.V., Korting, T.S., Fonseca, L.M.G., 2019. An Extensible and Easy-to-use Toolbox for Deep Learning Based Analysis of Remote Sensing Images, in: *IGARSS 2019 - 2019 IEEE International Geoscience and Remote Sensing Symposium*. IEEE, pp. 9815–9818. <https://doi.org/10.1109/IGARSS.2019.8898823>
- Maretto, R. V., Fonseca, L.M.G., Jacobs, N., Korting, T.S., Bendini, H.N., Parente, L.L., 2021. Spatio-Temporal Deep Learning Approach to Map Deforestation in Amazon Rainforest. *IEEE Geoscience and Remote Sensing Letters* 18, 771–775. <https://doi.org/10.1109/LGRS.2020.2986407>
- Martone, M., Rizzoli, P., Wecklich, C., González, C., Bueso-Bello, J.-L., Valdo, P., Schulze, D., Zink, M., Krieger, G., Moreira, A., 2018. The global forest/non-forest map from TanDEM-X interferometric SAR data. *Remote Sens Environ* 205, 352–373. <https://doi.org/10.1016/j.rse.2017.12.002>
- Matosak, B.M., Fonseca, L.M.G., Maretto, R.V., 2023. Deep Learning and Cloudy Optical Time Series: A Case of Study with LSTM to Map LULC in Pantanal, in: *International Geoscience and Remote Sensing Symposium (IGARSS)*. Institute of Electrical and Electronics Engineers Inc., pp. 7179–7182. <https://doi.org/10.1109/IGARSS52108.2023.10282993>
- Matosak, B.M., Fonseca, L.M.G., Taquary, E.C., Maretto, R.V., Bendini, H.D.N., Adami, M., 2022. Mapping Deforestation in Cerrado Based on Hybrid Deep Learning Architecture and Medium Spatial Resolution Satellite Time Series. *Remote Sens (Basel)* 14. <https://doi.org/10.3390/rs14010209>
- Maxwell, A.E., Warner, T.A., Fang, F., 2018. Implementation of machine-learning classification in remote sensing: an applied review. *Int J Remote Sens* 39, 2784–2817. <https://doi.org/10.1080/01431161.2018.1433343>
- Mermoz, S., Bouvet, A., Koleck, T., Ballère, M., Le Toan, T., 2021. Continuous Detection of Forest Loss in Vietnam, Laos, and Cambodia Using Sentinel-1 Data. *Remote Sens (Basel)* 13, 4877. <https://doi.org/10.3390/rs13234877>
- MMA, 2022. Prevenção e Controle do Desmatamento [WWW Document]. Ministério do Meio Ambiente. URL <http://combateaodesmatamento.mma.gov.br/> (accessed 12.22.22).
- Montavon, G., Orr, G.B., Müller, K.-R. (Eds.), 2012. *Neural Networks: Tricks of the Trade*, Lecture Notes in Computer Science. Springer Berlin Heidelberg, Berlin, Heidelberg. <https://doi.org/10.1007/978-3-642-35289-8>
- Mullissa, A., Reiche, J., Herold, M., 2023. Deep learning and automatic reference label harvesting for Sentinel-1 SAR-based rapid tropical dry forest disturbance mapping. *Remote Sens Environ* 298. <https://doi.org/10.1016/j.rse.2023.113799>
- Mullissa, A., Saatchi, S., Dalagnol, R., Erickson, T., Provost, N., Osborn, F., Ashary, A., Moon, V., Melling, D., 2024. LUCA: A Sentinel-1 SAR-Based Global Forest Land Use Change Alert. *Remote Sens (Basel)* 16, 2151. <https://doi.org/10.3390/rs16122151>
- Narvaes, I. da S., Santos, J.R. dos, Bispo, P. da C., Graça, P.M. de A., Guimarães, U.S., Gama, F.F., 2023. Estimating Forest Above-Ground Biomass in Central Amazonia Using Polarimetric Attributes of ALOS/PALSAR Images. *Forests* 14, 941. <https://doi.org/10.3390/f14050941>
- NICFI, N.I.C. and F.I., 2024. Satellite Data Programme [WWW Document]. NICFI.
- Nicolau, A.P., Flores-Anderson, A., Griffin, R., Herndon, K., Meyer, F.J., 2021. Assessing SAR C-band data to effectively distinguish modified land uses in a heavily disturbed Amazon forest.

- International Journal of Applied Earth Observation and Geoinformation 94, 102214.
<https://doi.org/10.1016/j.jag.2020.102214>
- Olofsson, P., Foody, G.M., Herold, M., Stehman, S. V., Woodcock, C.E., Wulder, M.A., 2014. Good practices for estimating area and assessing accuracy of land change. *Remote Sens Environ* 148, 42–57. <https://doi.org/10.1016/j.rse.2014.02.015>
- Ometto, J.P., Kalaba, K., Anshari, G.Z., Chacón, N., Farrell, A., Halim, S.A., Neufeldt, H., Sukumar, R., 2022. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, in: H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B.R. (Ed.), *Climate Change 2022: Impacts, Adaptation and Vulnerability*. Cambridge University Press, Cambridge, UK and New York, NY, USA, pp. 2369–2410.
<https://doi.org/10.1017/9781009325844.024>
- Ouchi, K., 2013. Recent trend and advance of synthetic aperture radar with selected topics. *Remote Sens (Basel)*.
- Paradella, W.R., Mura, J.C., Gama, F.F., Santos, A.R., 2012. Radar interferometry in surface deformation detection with orbital data. *Revista Brasileira de Cartografia* 64, 797–811.
- Parikh, H., Patel, S., Patel, V., 2020. Classification of SAR and PolSAR images using deep learning: a review. *Int J Image Data Fusion*. <https://doi.org/10.1080/19479832.2019.1655489>
- Passah, A., Sur, S.N., Paul, B., Kandar, D., 2022. SAR Image Classification: A Comprehensive Study and Analysis. *IEEE Access* 10, 20385–20399. <https://doi.org/10.1109/ACCESS.2022.3151089>
- Persson, H.J., Huuva, I., 2024. Polarimetric Measures in Biomass Change Prediction Using ALOS-2 PALSAR-2 Data. *Remote Sens (Basel)* 16, 953. <https://doi.org/10.3390/rs16060953>
- Planet Labs PBC, 2024. Planet Application Program Interface: In Space for Life on Earth [WWW Document]. NICFI Basemaps. URL <https://api.planet.com> (accessed 12.2.24).
- Pourshamsi, M., Xia, J., Yokoya, N., Garcia, M., Laval, M., Pottier, E., Balzter, H., 2021. Tropical forest canopy height estimation from combined polarimetric SAR and LiDAR using machine-learning. *ISPRS Journal of Photogrammetry and Remote Sensing* 172, 79–94.
<https://doi.org/10.1016/j.isprsjprs.2020.11.008>
- Quegan, S., Le Toan, T., Chave, J., Dall, J., Exbrayat, J.F., Minh, D.H.T., Lomas, M., D'Alessandro, M.M., Paillou, P., Papathanassiou, K., Rocca, F., Saatchi, S., Scipal, K., Shugart, H., Smallman, T.L., Soja, M.J., Tebaldini, S., Ulander, L., Villard, L., Williams, M., 2019. The European Space Agency BIOMASS mission: Measuring forest above-ground biomass from space. *Remote Sens Environ* 227, 44–60. <https://doi.org/10.1016/j.rse.2019.03.032>
- Quegan, S., Yu, J.J., 2001. Filtering of multichannel SAR images. *IEEE Transactions on Geoscience and Remote Sensing* 39, 2373–2379. <https://doi.org/10.1109/36.964973>
- Ramos, L.T., Sappa, A.D., 2025. Leveraging U-Net and selective feature extraction for land cover classification using remote sensing imagery. *Sci Rep* 15. <https://doi.org/10.1038/s41598-024-84795-1>
- Reiche, J., Hamunyela, E., Verbesselt, J., Hoekman, D., Herold, M., 2018. Improving near-real time deforestation monitoring in tropical dry forests by combining dense Sentinel-1 time series with Landsat and ALOS-2 PALSAR-2. *Remote Sens Environ* 204, 147–161.
<https://doi.org/10.1016/j.rse.2017.10.034>
- Reiche, J., Mullissa, A., Slagter, B., Gou, Y., Tsendbazar, N.-E., Odongo-Braun, C., Vollrath, A., Weisse, M.J., Stolle, F., Pickens, A., Donchyts, G., Clinton, N., Gorelick, N., Herold, M., 2021. Forest disturbance alerts for the Congo Basin using Sentinel-1. *Environmental Research Letters* 16, 024005. <https://doi.org/10.1088/1748-9326/abd0a8>
- Reis, M.S., Dutra, L.V., Sant'Anna, S.J.S., Escada, M.I.S., 2020. Multi-source change detection with PALSAR data in the Southern of Pará state in the Brazilian Amazon. *International Journal of Applied Earth Observation and Geoinformation* 84, 101945.
<https://doi.org/10.1016/j.jag.2019.101945>

- Ren, Y., Jiang, W., Liu, Y., 2023. A New Architecture of a Complex-Valued Convolutional Neural Network for PolSAR Image Classification. *Remote Sens (Basel)* 15, 4801. <https://doi.org/10.3390/rs15194801>
- Ronneberger, O., Fischer, P., Brox, T., 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation.
- Rostami, O., Kaveh, M., 2021. Optimal feature selection for SAR image classification using biogeography-based optimization (BBO), artificial bee colony (ABC) and support vector machine (SVM): a combined approach of optimization and machine learning. *Comput Geosci* 25, 911–930. <https://doi.org/10.1007/s10596-020-10030-1>
- SARMAP, 2016. SARscape 's Basic module Tutorial.
- Shafique, A., Cao, G., Khan, Z., Asad, M., Aslam, M., 2022. Deep Learning-Based Change Detection in Remote Sensing Images: A Review. *Remote Sens (Basel)*. <https://doi.org/10.3390/rs14040871>
- Shimada, M., Itoh, T., Motooka, T., Watanabe, M., Shiraishi, T., Thapa, R., Lucas, R., 2014. New global forest/non-forest maps from ALOS PALSAR data (2007–2010). *Remote Sens Environ* 155, 13–31. <https://doi.org/10.1016/j.rse.2014.04.014>
- Shuai, G., Zhang, J., Basso, B., Pan, Y., Zhu, X., Zhu, S., Liu, H., 2019. Multi-temporal RADARSAT-2 polarimetric SAR for maize mapping supported by segmentations from high-resolution optical image. *International Journal of Applied Earth Observation and Geoinformation* 74, 1–15. <https://doi.org/10.1016/j.jag.2018.08.021>
- Soja, M.J., Persson, H.J., Ulander, L.M.H., 2018. Modeling and Detection of Deforestation and Forest Growth in Multitemporal TanDEM-X Data. *IEEE J Sel Top Appl Earth Obs Remote Sens* 11, 3548–3563. <https://doi.org/10.1109/JSTARS.2018.2851030>
- Soja, M.J., Quegan, S., d'Alessandro, M.M., Banda, F., Scipal, K., Tebaldini, S., Ulander, L.M.H., 2021a. Mapping above-ground biomass in tropical forests with ground-cancelled P-band SAR and limited reference data. *Remote Sens Environ* 253. <https://doi.org/10.1016/j.rse.2020.112153>
- Soja, M.J., Quegan, S., d'Alessandro, M.M., Banda, F., Scipal, K., Tebaldini, S., Ulander, L.M.H., 2021b. Mapping above-ground biomass in tropical forests with ground-cancelled P-band SAR and limited reference data. *Remote Sens Environ* 253, 112153. <https://doi.org/10.1016/j.rse.2020.112153>
- Solórzano, J. V., Mas, J.F., Gallardo-Cruz, J.A., Gao, Y., Fernández-Montes de Oca, A., 2023. Deforestation detection using a spatio-temporal deep learning approach with synthetic aperture radar and multispectral images. *ISPRS Journal of Photogrammetry and Remote Sensing* 199, 87–101. <https://doi.org/10.1016/j.isprsjprs.2023.03.017>
- Solórzano, J. V., Mas, J.F., Gao, Y., Gallardo-Cruz, J.A., 2021. Land use land cover classification with U-net: Advantages of combining sentinel-1 and sentinel-2 imagery. *Remote Sens (Basel)* 13. <https://doi.org/10.3390/rs13183600>
- Song, R., Wang, W., Yu, W., 2024. The Latest Developments in Spaceborne High-Resolution Wide-Swath SAR Systems and Imaging Methods. *Sensors*. <https://doi.org/10.3390/s24185978>
- Song, X.-P., Hansen, M.C., Stehman, S. V., Potapov, P. V., Tyukavina, A., Vermote, E.F., Townshend, J.R., 2018. Global land change from 1982 to 2016. *Nature* 560, 639–643. <https://doi.org/10.1038/s41586-018-0411-9>
- Souza, C.M., Z. Shimbo, J., Rosa, M.R., Parente, L.L., A. Alencar, A., Rudorff, B.F.T., Hasenack, H., Matsumoto, M., G. Ferreira, L., Souza-Filho, P.W.M., de Oliveira, S.W., Rocha, W.F., Fonseca, A. V., Marques, C.B., Diniz, C.G., Costa, D., Monteiro, D., Rosa, E.R., Vélez-Martin, E., Weber, E.J., Lenti, F.E.B., Paternost, F.F., Pareyn, F.G.C., Siqueira, J. V., Viera, J.L., Neto, L.C.F., Saraiva, M.M., Sales, M.H., Salgado, M.P.G., Vasconcelos, R., Galano, S., Mesquita, V. V., Azevedo, T., 2020. Reconstructing Three Decades of Land Use and Land Cover Changes in Brazilian Biomes with Landsat Archive and Earth Engine. *Remote Sens (Basel)* 12, 2735. <https://doi.org/10.3390/rs12172735>
- Strand, J., Soares-Filho, B., Costa, M.H., Oliveira, U., Ribeiro, S.C., Pires, G.F., Oliveira, A., Rajão, R., May, P., van der Hoff, R., Siikamäki, J., da Motta, R.S., Toman, M., 2018. Spatially explicit

- valuation of the Brazilian Amazon Forest's Ecosystem Services. *Nat Sustain* 1, 657–664. <https://doi.org/10.1038/s41893-018-0175-0>
- Sugimoto, R., Kato, S., Nakamura, R., Tsutsumi, C., Yamaguchi, Y., 2022. Deforestation detection using scattering power decomposition and optimal averaging of volume scattering power in tropical rainforest regions. *Remote Sens Environ* 275, 113018. <https://doi.org/10.1016/j.rse.2022.113018>
- Sullivan, R.J., 2004. Radar Foundations for Imaging and Advanced Concepts. Institution of Engineering and Technology. <https://doi.org/10.1049/SBRA030E>
- Tadesse, G.A., Robinson, C., Mwangi, C., Maina, E., Nyakundi, J., Marotti, L., Hacheme, G.Q., Alemohammad, H., Dodhia, R., Ferres, J.M.L., 2024. Local vs. Global: Local Land-Use and Land-Cover Models Deliver Higher Quality Maps.
- Tavares, P., Beltrão, N., Guimarães, U., Teodoro, A., 2019. Integration of Sentinel-1 and Sentinel-2 for Classification and LULC Mapping in the Urban Area of Belém, Eastern Brazilian Amazon. *Sensors* 19, 1140. <https://doi.org/10.3390/s19051140>
- Taylor, K.E., 2001. Summarizing multiple aspects of model performance in a single diagram. *Journal of Geophysical Research: Atmospheres* 106, 7183–7192. <https://doi.org/10.1029/2000JD900719>
- Tuia, D., Schindler, K., Demir, B., Zhu, X.X., Kochupillai, M., Džeroski, S., van Rijn, J.N., Hoos, H.H., Del Frate, F., Datcu, M., Markl, V., Le Saux, B., Schneider, R., Camps-Valls, G., 2025. Artificial Intelligence to Advance Earth Observation: A review of models, recent trends, and pathways forward. *IEEE Geosci Remote Sens Mag* 2–25. <https://doi.org/10.1109/MGRS.2024.3425961>
- USGS, (United States Geology Survey), 2020. EarthExplorer - USGS [WWW Document]. Earth Explorer. URL <http://earthexplorer.usgs.gov/> (accessed 4.4.15).
- Vaglio Laurin, G., Liesenberg, V., Chen, Q., Guerriero, L., Del Frate, F., Bartolini, A., Coomes, D., Wilebore, B., Lindsell, J., Valentini, R., 2013. Optical and SAR sensor synergies for forest and land cover mapping in a tropical site in West Africa. *International Journal of Applied Earth Observation and Geoinformation* 21, 7–16. <https://doi.org/10.1016/j.jag.2012.08.002>
- van Der Sanden, J.J., 1997. Radar Remote Sensing To Support Tropical Forest Management, Tropenbos-Guyana Series 5.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention Is All You Need.
- Wagner, F.H., Dalagnol, R., Silva-Junior, C.H.L., Carter, G., Ritz, A.L., Hirye, M.C.M., Ometto, J.P.H.B., Saatchi, S., 2023. Mapping Tropical Forest Cover and Deforestation with Planet NICFI Satellite Images and Deep Learning in Mato Grosso State (Brazil) from 2015 to 2021. *Remote Sens (Basel)* 15. <https://doi.org/10.3390/rs15020521>
- Wang, G., Li, B., Zhang, T., Zhang, S., 2022. A Network Combining a Transformer and a Convolutional Neural Network for Remote Sensing Image Change Detection. *Remote Sens (Basel)* 14. <https://doi.org/10.3390/rs14092228>
- Wang, J., Yang, Y., Mao, J., Huang, Z., Huang, C., Xu, W., 2016. CNN-RNN: A Unified Framework for Multi-label Image Classification.
- Wang, L., Zhang, C., Li, R., Duan, C., Meng, X., Atkinson, P.M., 2021. Scale-aware Neural Network for Semantic Segmentation of Multi-resolution Remote Sensing Images.
- Watanabe, M., Koyama, C.N., Hayashi, M., Nagatani, I., Shimada, M., 2018. Early-Stage Deforestation Detection in the Tropics With L-band SAR. *IEEE J Sel Top Appl Earth Obs Remote Sens* 11, 2127–2133. <https://doi.org/10.1109/JSTARS.2018.2810857>
- Watanabe, M., Koyama, C.N., Hayashi, M., Nagatani, I., Tadono, T., Shimada, M., 2021. Refined algorithm for forest early warning system with ALOS-2/PALSAR-2 ScanSAR data in tropical forest regions. *Remote Sens Environ* 265, 112643. <https://doi.org/10.1016/j.rse.2021.112643>
- Whyte, A., Ferentinos, K.P., Petropoulos, G.P., 2018. A new synergistic approach for monitoring wetlands using Sentinels -1 and 2 data with object-based machine learning algorithms. *Environmental Modelling & Software* 104, 40–54. <https://doi.org/10.1016/j.envsoft.2018.01.023>

- Woodhouse, I.H., 2017. Introduction to Microwave Remote Sensing. CRC Press, New York.
<https://doi.org/10.1201/9781315272573>
- Yu, F., Koltun, V., 2015. Multi-Scale Context Aggregation by Dilated Convolutions.
- Yu, T., Wu, W., Gong, C., Li, X., 2021. Residual multi-attention classification network for a forest dominated tropical landscape using high-resolution remote sensing imagery. *ISPRS Int J Geoinf* 10. <https://doi.org/10.3390/ijgi10010022>
- Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.-E., Xu, P., Ramoino, F., Arino, O., 2022. ESA WorldCover 10 m 2021 v200.
<https://doi.org/https://doi.org/10.5281/zenodo.7254221>
- Zhang, X., Liu, L., Chen, X., Gao, Y., Xie, S., Mi, J., 2021. GLC_FCS30: global land-cover product with fine classification system at 30 m using time-series Landsat imagery. *Earth Syst Sci Data* 13, 2753–2776. <https://doi.org/10.5194/essd-13-2753-2021>
- Zhao, F., Sun, R., Zhong, L., Meng, R., Huang, C., Zeng, X., Wang, M., Li, Y., Wang, Z., 2022. Monthly mapping of forest harvesting using dense time series Sentinel-1 SAR imagery and deep learning. *Remote Sens Environ* 269, 112822. <https://doi.org/10.1016/j.rse.2021.112822>
- Zhou, Y., Wang, H., Xu, F., Jin, Y.Q., 2016. Polarimetric SAR Image Classification Using Deep Convolutional Neural Networks. *IEEE Geoscience and Remote Sensing Letters* 13, 1935–1939. <https://doi.org/10.1109/LGRS.2016.2618840>
- Zhu, Q., Guo, X., Li, Z., Li, D., 2024. A review of multi-class change detection for satellite remote sensing imagery. *Geo-spatial Information Science* 27, 1–15.
<https://doi.org/10.1080/10095020.2022.2128902>
- Zhu, Y., Geis, C., So, E., Jin, Y., 2021. Multitemporal Relearning with Convolutional LSTM Models for Land Use Classification. *IEEE J Sel Top Appl Earth Obs Remote Sens* 14, 3251–3265.
<https://doi.org/10.1109/JSTARS.2021.3055784>